

Pricing Exotic Options using Monte Carlo Techniques

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Abstract

We present a Bayesian calibration of exotic option pricing models using both traditional Markov chain Monte Carlo and Hamiltonian Monte Carlo samplers in a unified framework. In our study, an Asian option is priced under two alternative dynamics, a Merton jump-diffusion model and a Heston stochastic volatility model, while a Margrabe exchange option is priced under a two-asset Black-Scholes model. Full Bayesian inference is performed for all model parameters, and model fit is evaluated via posterior predictive checks, Greek sensitivity analyses, residual error diagnostics, and price surface comparisons to a benchmark model. The HMC algorithm is applied to efficiently sample the posterior for the continuous parameter models, whereas adaptive Metropolis-within-Gibbs and birth-death Metropolis-within-Gibbs are applied for the non-HMC calibrations. The resulting posterior distributions reveal that the Heston model calibrated with HMC provides a better fit to the Asian option data over the Merton jump-diffusion model, capturing some of the implied-volatility skew and yielding better credible intervals for model prices compared to the Merton jump-diffusion model. The Merton jump-diffusion model exhibits wide posterior uncertainty bands, reflecting greater uncertainty in parameter estimates. For the Margrabe exchange option, the simple two-asset Black-Scholes model achieves a near-perfect fit under both non-HMC and HMC samplers, indicating little benefit from additional model complexity in that case. Model selection criteria (Bayesian and Deviance Information Criteria) further support these findings, with the Heston model being preferred over the Merton jump-diffusion model for the Asian option, while the Black-Scholes model is confirmed to be sufficient for the exchange option. These results show the effectiveness of Bayesian calibration for less complex models such as a multi-asset Black-Scholes in capturing essential market features and quantifying uncertainty in exotic option pricing models.

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1 Introduction

Quantitative finance and algorithmic trading have expanded dramatically over the last two decades. Automated, model-driven strategies now dominate many markets. For example, an estimated 70% of U.S. equity trading volume is initiated through algorithmic trading (Yadav, 2015). Global algorithmic trading activity continues to grow at double-digit rates annually, increasing reliance on complex models. Options and other derivatives markets have similarly surged in activity. Cboe Global Markets reported a record 3.8 billion options contracts traded in 2024, averaging 14.95 million contracts daily, marking the fifth consecutive year of record-breaking volume.[1]

Quantitative hedge funds and trading firms that leverage these models have achieved remarkable success. Industry leaders such as Renaissance Technologies and Marshall Wace delivered double-digit returns in 2024, outperforming many traditional asset managers. In particular, Marshall Wace's Eureka fund earned approximately 14.3%, its Market Neutral Tops fund returned 22.6%, and its Alpha Plus fund gained 15.9% in 2024.[2] The scale of assets under management at top quantitative firms has also grown significantly. Citadel LLC manages approximately $397 billion as of August 2024,[3] and its market-making arm, Citadel Securities, reportedly generated a record $9.7 billion in trading revenue in 2024, a 55% increase from the previous year.[4] Such figures reflect not only impressive performance, but also the breadth of algorithmic option trading in modern markets. In particular, leading firms have built sophisticated option pricing engines, usually based on stochastic models, that seek to exploit small pricing inefficiencies at large scale. This fusion of mathematical finance and computation has been a key driver behind the success of quantitative trading firms and the efficiency of derivative markets.

Despite these advances, traditional approaches to option pricing and model calibration face significant challenges when confronted with the complexity of real markets. Classical parameter estimation techniques, such as maximum likelihood estimation or least-squares calibration to market option prices, encounter both computational and statistical difficulties. Complex option pricing models with stochastic volatility or jump processes often have likelihood functions that require integrating over unobserved state variables, such as latent volatility paths or latent jump processes. These integrations quickly become impractical in high dimensions (Brouillon et al., 2024). As the number of model parameters grows, conventional optimisation methods also struggle. Likelihood surfaces can be highly non-convex, with multiple local optima, and convergence to a meaningful global solution can be difficult to guarantee. Moreover, classical calibration typically produces point estimates that do not measure uncertainty in the parameters. This is problematic in risk management and inference, where understanding the range of plausible parameter values and their dependence structure is often as important as finding a single best-fit value.

Similar issues arise on the pricing side with classical deterministic techniques. Partial

[1] https://ir.cboe.com/news/news-details/2025/Cboe-Global-Markets-Reports-Trading-Volume-for-December-and-Full-Year-2024/

[2] https://www.reuters.com/markets/hedge-funds-deliver-double-digit-returns-2024-2025-01-02/#:~:text=British%20hedge%20fund%20Marshall%20Wace,Eureka%20fund%2C%20a%20source%20said.

[3] https://www.investopedia.com/articles/personal-finance/011515/worlds-top-10-hedge-fund-firms.asp

[4] https://www.bloomberg.com/news/articles/2025-03-06/citadel-securities-9-7-billion-trading-revenue-passes-barclays

differential equation solvers and lattice methods, such as binomial or trinomial trees, provide elegant solutions for simple option models, but they suffer from the curse of dimensionality. The dimension of a pricing problem grows with the number of underlying risk factors, latent state variables, and path-dependent quantities. Standard grid-based methods therefore become infeasible beyond a small number of dimensions (Han et al., 2018). This is especially relevant for exotic options. An Asian option depends on an average of the underlying path, while a multi-asset option such as a Margrabe exchange option depends on the joint behaviour of several assets. Although closed-form pricing formulas exist in certain idealised cases, such as Black-Scholes European options, many exotic options have no simple analytical solution. Traders and researchers often turn to Monte Carlo simulation for such cases, but straightforward Monte Carlo can converge slowly, and embedding it inside a calibration loop significantly increases the computational load.

These limitations motivate the use of more flexible, simulation-based approaches. Bayesian methods powered by *Markov chain Monte Carlo (MCMC)* provide an alternative for inference in quantitative finance. In a Bayesian setting, model parameters are treated as random variables with prior distributions, and these are updated to posterior distributions after observing data. MCMC algorithms are then used to sample from the posterior distribution. This changes the nature of the calibration problem. Instead of producing only a single calibrated parameter vector, Bayesian calibration produces a distribution over plausible parameter values. This allows parameter uncertainty, posterior dependence, and weak identification to be studied directly.

The fundamental MCMC technique is the *Metropolis-Hastings* algorithm, which can be adapted to many posterior distributions because it only requires the target density up to a normalising constant. It is often combined with Gibbs-style blocking when different groups of parameters have different scales or dependence structures. Adaptive Metropolis-Hastings improves on a fixed random-walk proposal by learning the posterior covariance structure during the simulation. This is useful when the posterior contains strong correlation between parameters, as occurs in multi-asset option calibration. *Birth-death MCMC* and related reversible-jump methods are also useful when the number of latent variables is unknown, as in jump-diffusion models where the number of jumps is itself random. Hamiltonian Monte Carlo is another important development. It uses gradient information from the posterior density to propose longer and more directed moves, reducing the random-walk behaviour of traditional MCMC (Hoffman et al., 2014). These methods form a flexible toolkit for Bayesian inference in financial models with nonlinear pricing maps and complex posterior geometry.

In this paper, we use these ideas to study Bayesian MCMC calibration for exotic option pricing. The paper focuses on two exotic payoff structures. The first is the geometric Asian option, which is path-dependent because its payoff depends on an average of the underlying asset path. The second is the Margrabe exchange option, which is a multi-asset option giving the holder the right to exchange one asset for another at maturity. These two examples are deliberately chosen because they expose different forms of complexity. The Asian option highlights path-dependence and the difficulty of high-dimensional path simulation. The Margrabe option highlights multi-asset dependence and the difficulty of identifying volatility and correlation parameters from option prices.

We consider three stochastic models. The Black-Scholes model is used as an analytically

tractable benchmark. Under Black-Scholes dynamics, the geometric Asian option admits a closed-form price because the logarithm of the geometric average is normally distributed. The Margrabe exchange option also admits a closed-form expression under a two-asset Black-Scholes model through a change of numeraire. The Merton jump-diffusion model is then introduced to capture discontinuous price movements and heavier-tailed return distributions. The Heston stochastic volatility model is introduced to capture time-varying variance and leverage effects. For the Merton and Heston Asian option experiments, model-implied prices are computed using Monte Carlo simulation.

The paper is organised as follows. Section 2 introduces the option payoffs and stochastic asset models used throughout the paper. It defines the geometric Asian option and the Margrabe exchange option, before presenting the Black-Scholes, Merton jump-diffusion, and Heston stochastic volatility dynamics under the risk-neutral measure. Section 3 derives the analytical benchmark prices for the Black-Scholes geometric Asian option and the Black-Scholes Margrabe exchange option. It then explains how Monte Carlo simulation is used when closed-form prices are unavailable, particularly for path-dependent payoffs under Merton and Heston dynamics. Section 4 develops the Bayesian calibration framework and introduces the MCMC methods used in the numerical experiments, including Metropolis-Hastings, *adaptive Metropolis-Hastings, Metropolis-Hastings within Gibbs*, birth-death MCMC, and *Hamiltonian Monte Carlo*.

Section 5 presents the numerical experiments. The experiments are synthetic rather than based on observed exotic option market data, because reliable historical data for exotic options is difficult to obtain, especially for path-dependent over-the-counter contracts with non-standard monitoring conventions. The experiments begin with validation cases based on the Black-Scholes geometric Asian and Margrabe formulas. They then study correctly specified Merton and Heston geometric Asian option calibrations. Finally, a misspecification experiment fits a Merton jump-diffusion model to Heston-generated Asian option prices in order to test whether posterior predictive diagnostics and residual heatmaps reveal structural model error.

2 Option Models and Exotic Payoffs

The purpose of this section is to introduce the financial models and option payoffs used throughout the paper. Since we are focused on Bayesian calibration and MCMC methods for pricing exotic options, we require a stochastic model for the underlying asset, a payoff function for the exotic derivative contract, and pricing rules that assign the contract a no-arbitrage value.

We work throughout in continuous time on a finite time interval $[0, T]$, where $T > 0$ is the maturity of the option. The models considered in this paper are all specified under a risk-neutral probability measure $\mathbb{Q}$. Under this measure, discounted tradable asset prices are martingales, which is the standard setting for no-arbitrage derivative valuation.

Let $(\Omega, \mathcal{F}, \mathbb{Q})$ be a filtered probability space satisfying the usual conditions. The filtration $\{\mathcal{F}_t\}_{t\geq 0}$ represents the information available up to time t, and we assume the existence of a risk-free asset B_t, interpreted as a money-market account, satisfying

$$dB_t = rB_t dt, \qquad B_t = e^{rt}, \qquad B_0 = 1,$$

where $r \geq 0$ is a constant risk-free rate. For a contingent claim with maturity payoff H, provided H is integrable under $\mathbb{Q}$, the time-zero no-arbitrage price is $C_0 = e^{-rT}\,\mathbb{E}_{\mathbb{Q}}[H]$. More generally, the time-t price is

$$C_t = e^{-r(T-t)}\,\mathbb{E}_{\mathbb{Q}}\left[H \mid \mathcal{F}_t\right], \qquad 0 \leq t \leq T.$$

Let $\boldsymbol{\theta}$ denote the vector of model parameters. Given a payoff functional g, the model-implied price associated with parameter vector $\boldsymbol{\theta}$ can be written as

$$C(\boldsymbol{\theta}) = e^{-rT}\,\mathbb{E}_{\mathbb{Q},\boldsymbol{\theta}}\left[g\right].$$

For a vanilla European call option written on a single underlying asset S, the payoff depends only on the terminal price S_T:

$$g(S_T) = (S_T - K)^+,$$

where K is the strike and $x^+ = \max(0, x)$. In contrast, an exotic option may depend on several prices along the path, on an average of prices, on a maximum or minimum, on whether the asset has crossed a barrier, or on the relative performance of several assets. In such cases the payoff is better written as a functional of the path, i.e.,$g = g\left(\{S_t\}_{0\leq t\leq T}\right)$. For a multi-asset option written on d underlying assets $\boldsymbol{S}_t = (S_{1,t}, \ldots, S_{d,t})$, the payoff may instead be a functional of the vector-valued path, in other words, $g = g\left(\{\boldsymbol{S}_t\}_{0\leq t\leq T}\right)$. The price is the discounted expectation of the payoff under the stochastic model. In this paper, the same payoff will be priced under different risk-neutral models, and the same model will be calibrated using different MCMC methods.

2.1 Exotic Options

In this paper, we consider the *Asian option* and the *Margrabe option* as our exotic options. Asian options are path-dependent derivatives whose payoff depends on an average of the underlying asset price over the life of the option. Let $S = \{S_t\}_{0\leq t\leq T}$ be the price process of

the underlying asset. There are two common forms of the Asian averages: the arithmetic average and the geometric average. In continuous time, the arithmetic average is

$$A_T^{\text{arith}} = \frac{1}{T}\int_0^T S_t dt.$$

The corresponding time-zero risk-neutral price of the arithmetic Asian call is:

$$C_A^{\text{arith}} = e^{-rT}\,\mathbb{E}_{\mathbb{Q}}\left[\left(\frac{1}{T}\int_0^T S_t dt - K\right)^+\right].$$

The alternative Asian payoff uses the geometric average, which is defined by averaging the logarithm of the asset price and then exponentiating:

$$A_T^{\text{geom}} = \exp\left(\frac{1}{T}\int_0^T \log S_t dt\right),$$

and its time-zero price for the Asian call payoff is

$$C_A^{\text{geom}} = e^{-rT}\,\mathbb{E}_{\mathbb{Q}}\left[\left(\exp\left(\frac{1}{T}\int_0^T \log S_t dt\right) - K\right)^+\right].$$

In practice, Asian options are often monitored at a finite collection of dates rather than continuously. Let $0 < t_1 < t_2 < \cdots < t_m = T$ be the monitoring dates. The discretely monitored arithmetic and geometric average call prices are

$$C_{A,m}^{\text{arith}} = e^{-rT}\mathbb{E}^{\mathbb{Q}}\left[\left(\frac{1}{m}\sum_{j=1}^m S_{t_j} - K\right)^+\right], \qquad C_{A,m}^{\text{geom}} = e^{-rT}\mathbb{E}^{\mathbb{Q}}\left[\left(\exp\left(\frac{1}{m}\sum_{j=1}^m \log S_{t_j}\right) - K\right)^+\right]$$

The arithmetic Asian option is usually more relevant in applications because the arithmetic average is the natural average of prices. However, even under the Black-Scholes model, the arithmetic average of lognormal random variables is not itself lognormally distributed, hence, a simple closed-form pricing formula is not available. The geometric Asian option is mathematically more tractable because the logarithm of the geometric average is a linear functional of the log-price process. Under Black-Scholes dynamics, this implies that the logarithm of the geometric average is normally distributed. Consequently, the geometric Asian option gives us an analytical formula under Black-Scholes, which will be derived in Section 3. This makes the geometric Asian option a useful benchmark for validating numerical methods and will be our choice for the Asian payoff for this paper.

For Asian options, if the option is monitored at m dates, then the payoff depends on the random vector $(S_{t_1}, S_{t_2}, \ldots, S_{t_m})$. One must then either augment the state variable with the accumulated average or solve a higher-dimensional problem using a grid-based partial differential equation method. As m increases, or as the stochastic model itself becomes more complex, deterministic methods can become computationally expensive. Later in the paper, we discuss how Monte Carlo methods help address this problem.

The second exotic payoff considered in this paper is the Margrabe exchange option. A Margrabe option gives the holder the right, but not the obligation, to exchange one asset

for another at maturity. Let $S_{1,t}$ and $S_{2,t}$ denote the prices of two risky assets. The price at time-zero for the payoff of the option at maturity T is

$$C_M = e^{-rT}\mathbb{E}_{\mathbb{Q}}\left[(S_{1,T} - S_{2,T})^+\right].$$

Unlike the Asian, the Margrabe is not path-dependent, since the payoff depends only on the terminal values of the two assets. The option is sensitive to both volatilities of the two assets and their correlation. Conveniently, if we rewrite this payoff as

$$(S_{1,T} - S_{2,T})^+ = S_{2,T}\left(\frac{S_{1,T}}{S_{2,T}} - 1\right)^+,$$

then under a two-asset Black-Scholes model, we have the option to use $S_{2,t}$ as the numeraire. Under this change of numeraire, the ratio $S_{1,t}/S_{2,t}$ becomes the relevant tradable process, and the option resembles a European call on the exchange ratio with unit strike. hence, this European call depends on the *effective exchange volatility*

$$\sigma_{\text{ex}} = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2},$$

where σ_1 and σ_2 are the volatilities of S_1 and S_2, and ρ is the instantaneous correlation between their Brownian shocks.

2.2 Stochastic Modelling

2.2.1 Black-Scholes Dynamics

The Black-Scholes model is the simplest continuous-time model considered in this paper. Although the Black-Scholes model is too restrictive to capture many empirical features of asset returns, it remains widely used because of its analytical tractability. In particular, both the geometric Asian option and the Margrabe exchange option admit closed-form expressions under Black-Scholes-type assumptions. For a single asset, the risk-neutral Black-Scholes dynamics are

$$\frac{dS_t}{S_t} = r\,dt + \sigma\,dW_t, \qquad S_0 > 0, \tag{1}$$

where $\sigma > 0$ is the volatility and W_t is a standard Brownian motion under $\mathbb{Q}$. Equivalently, under the risk-neutral measure,

$$dS_t = rS_t\,dt + \sigma S_t\,dW_t.$$

Applying Itô's lemma to $X_t = \log S_t$ gives

$$dX_t = \frac{1}{S_t}dS_t - \frac{1}{2S_t^2}(dS_t)^2,$$

which we can rearrange to be

$$dX_t = \left(r - \frac{1}{2}\sigma^2\right)dt + \sigma\,dW_t.$$

Integrating dX_t from 0 to t yields

$$X_t = X_0 + \left(r - \frac{1}{2}\sigma^2\right)t + \sigma W_t.$$

Since $X_0 = \log S_0$, the asset price is

$$S_t = S_0 \exp\left[\left(r - \frac{1}{2}\sigma^2\right)t + \sigma W_t\right].$$

Thus $S_t > 0$ almost surely for all $t \geq 0$, and

$$\log S_t \sim \mathcal{N}\left(\log S_0 + \left(r - \frac{1}{2}\sigma^2\right)t, \sigma^2 t\right).$$

Since the exponential martingale has expectation one, it follows that $\mathbb{E}^{\mathbb{Q}}[S_t] = S_0 e^{rt}$, which is consistent with risk-neutral valuation.

For the Margrabe option, we require a two-asset version of the Black-Scholes model. Let $W_{1,t}$ and $W_{2,t}$ be Brownian motions under $\mathbb{Q}$ satisfying

$$d\langle W_1, W_2\rangle_t = \rho\, dt, \qquad \rho \in (-1, 1).$$

The two assets satisfy

$$\frac{dS_{1,t}}{S_{1,t}} = r\, dt + \sigma_1\, dW_{1,t}, \qquad \frac{dS_{2,t}}{S_{2,t}} = r\, dt + \sigma_2\, dW_{2,t},$$

where $\sigma_1 > 0$ and $\sigma_2 > 0$. By applying Itô's lemma to each log-price,

$$d\log S_{i,t} = \left(r - \frac{1}{2}\sigma_i^2\right)dt + \sigma_i\, dW_{i,t}, \qquad i = 1, 2.$$

Therefore, the log exchange ratio

$$R_t = \left(\frac{S_{1,t}}{S_{2,t}}\right), \qquad X_t = \log R_t = \log S_{1,t} - \log S_{2,t}.$$

satisfying,

$$\begin{aligned} dX_t &= d\log S_{1,t} - d\log S_{2,t} \\ &= \left(r - \frac{1}{2}\sigma_1^2\right)dt - \left(r - \frac{1}{2}\sigma_2^2\right)dt + \sigma_1\, dW_{1,t} - \sigma_2\, dW_{2,t} \\ &= -\frac{1}{2}(\sigma_1^2 - \sigma_2^2)dt + \sigma_1\, dW_{1,t} - \sigma_2\, dW_{2,t}. \end{aligned}$$

The variance of the Brownian increment in the exchange ratio is

$$\begin{aligned} \mathbb{V}\mathrm{ar}^{\mathbb{Q}}\left(\sigma_1\, dW_{1,t} - \sigma_2\, dW_{2,t}\right) &= \sigma_1^2\, dt + \sigma_2^2\, dt - 2\sigma_1\sigma_2\, d\langle W_1, W_2\rangle_t \\ &= \left(\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2\right)dt. \end{aligned}$$

As mentioned in the previous subsection, we can define the exchange volatility as $\sigma_{\text{ex}} = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$.

2.2.2 Merton Jump-Diffusion Dynamics

One assumption of the Black-Scholes model is that log-returns are normally distributed and that asset prices move continuously. Empirically, financial returns often exhibit heavier tails and sudden large movements. The Merton jump-diffusion model extends

Black-Scholes by adding a compound Poisson jump component to the asset price dynamics. This allows the model to represent discontinuous price changes while retaining a relatively tractable structure.

Under the risk-neutral measure $\mathbb{Q}$, the Merton jump-diffusion model is specified by

$$\frac{dS_t}{S_{t^-}} = (r - \lambda\kappa_J)\,dt + \sigma\,dW_t + (J-1)\,dN_t. \tag{2}$$

Here S_{t^-} denotes the left limit of S at time t, W_t is a Brownian motion under $\mathbb{Q}$, and N_t is a Poisson process with intensity $\lambda \geq 0$. The random variable $J > 0$ is the multiplicative jump size. When a jump occurs, the asset price changes from S_{t^-} to $S_t = S_{t^-}J$. We write $J = e^Y$, where the log-jump size Y is normally distributed:

$$Y \sim \mathcal{N}(\nu, \delta^2).$$

The jump-size compensation term is

$$\kappa_J = \mathbb{E}^{\mathbb{Q}}[J-1] = \mathbb{E}^{\mathbb{Q}}[e^Y - 1].$$

Since $Y \sim \mathcal{N}(\nu, \delta^2)$, its moment generating function gives

$$\mathbb{E}^{\mathbb{Q}}[e^Y] = \exp\left(\nu + \frac{1}{2}\delta^2\right).$$

Hence $\kappa_J = \exp\left(\nu + \frac{1}{2}\delta^2\right) - 1$. The drift in (2) is $r - \lambda\kappa_J$ rather than r. This compensation ensures that the discounted asset price is a martingale under $\mathbb{Q}$. To see this explicitly, the solution of the Merton model can be written as

$$S_t = S_0 \exp\left[\left(r - \lambda\kappa_J - \frac{1}{2}\sigma^2\right)t + \sigma W_t + \sum_{i=1}^{N_t} Y_i\right], \tag{3}$$

where $\{Y_i\}_{i\geq 1}$ are independent copies of Y, independent of W_t and N_t. Taking expectations,

$$\mathbb{E}^{\mathbb{Q}}[S_t] = S_0 \exp\left[\left(r - \lambda\kappa_J - \frac{1}{2}\sigma^2\right)t\right] \mathbb{E}^{\mathbb{Q}}\left[e^{\sigma W_t}\right] \mathbb{E}^{\mathbb{Q}}\left[\exp\left(\sum_{i=1}^{N_t} Y_i\right)\right].$$

For the compound Poisson term, conditioning on $N_t = n$ gives

$$\mathbb{E}^{\mathbb{Q}}\left[\exp\left(\sum_{i=1}^{N_t} Y_i\right) \mid N_t = n\right] = \left(\mathbb{E}^{\mathbb{Q}}[e^Y]\right)^n.$$

Therefore,

$$\begin{aligned}
\mathbb{E}^{\mathbb{Q}}\left[\exp\left(\sum_{i=1}^{N_t} Y_i\right)\right] &= \sum_{n=0}^{\infty} \left(\mathbb{E}^{\mathbb{Q}}[e^Y]\right)^n e^{-\lambda t}\frac{(\lambda t)^n}{n!} \\
&= \exp(-\lambda t)\sum_{n=0}^{\infty} \frac{\left(\lambda t\,\mathbb{E}^{\mathbb{Q}}[e^Y]\right)^n}{n!} \\
&= \exp\left[\lambda t\left(\mathbb{E}^{\mathbb{Q}}[e^Y] - 1\right)\right] \\
&= \exp(\lambda\kappa_J t).
\end{aligned}$$

Combining these terms gives

$$\begin{aligned}\mathbb{E}^{\mathbb{Q}}[S_t] &= S_0 \exp\left[\left(r - \lambda\kappa_J - \frac{1}{2}\sigma^2\right)t\right] \exp\left(\frac{1}{2}\sigma^2 t\right) \exp(\lambda\kappa_J t) \\ &= S_0 e^{rt}.\end{aligned}$$

Hence $e^{-rt}S_t$ has constant expectation, which is necessary for our risk-neutral measure. Now, by taking logarithms in (3), we obtain

$$\log S_t = \log S_0 + \left(r - \lambda\kappa_J - \frac{1}{2}\sigma^2\right)t + \sigma W_t + \sum_{i=1}^{N_t} Y_i.$$

Conditioning on $N_t = n$, the jump sum is normally distributed:

$$\sum_{i=1}^{n} Y_i \sim \mathcal{N}(n\nu, n\delta^2).$$

Thus,

$$\log S_t \mid N_t = n \sim \mathcal{N}\left(\log S_0 + \left(r - \lambda\kappa_J - \frac{1}{2}\sigma^2\right)t + n\nu, \sigma^2 t + n\delta^2\right).$$

Unconditionally, $\log S_t$ is not normally distributed. It is a Poisson mixture of normal distributions. This distinction is important. Path-dependent averages under the Merton model generally do not remain lognormal after integrating over random jump times and jump sizes; hence, finding analytical solutions is more challenging. For this reason, in this paper the Merton model is treated primarily as a model requiring Monte Carlo pricing for exotic payoffs such as Asian options.

The parameter vector for the one-dimensional Merton jump-diffusion model is written as $\boldsymbol{\theta}_{\text{MJD}} = (\sigma, \lambda, \nu, \delta)$, with constraints

$$\sigma > 0, \qquad \lambda \geq 0, \qquad \nu \in \mathbb{R}, \qquad \delta \geq 0.$$

2.2.3 Heston Dynamics

The Heston model extends Black-Scholes by allowing the instantaneous variance of the asset return to evolve randomly over time. Empirically, many assets exhibit volatility clustering, where periods of high volatility tend to be followed by further periods of high volatility, and periods of low volatility tend to be followed by further periods of low volatility. Hence, a constant-volatility model cannot capture this behaviour. Under the risk-neutral measure, the Heston model is

$$\begin{aligned}\frac{dS_t}{S_t} &= r\,dt + \sqrt{v_t}\,dW_{S,t}, \\ dv_t &= \kappa(\xi - v_t)\,dt + \sigma\sqrt{v_t}\,dW_{v,t},\end{aligned}$$

where the correlation satisfies $d\langle W_S, W_v\rangle_t = \rho\,dt$ with $\rho \in (-1, 1)$. The variance process v_t follows a *Cox–Ingersoll–Ross square-root diffusion*. The parameter $\kappa > 0$ controls the speed of mean reversion, $\xi > 0$ is the long-run variance level, and $\sigma > 0$ is the volatility

of volatility. The initial variance is assumed to be positive. The drift term $\kappa(\xi - v_t)$ pulls the variance process back toward ξ.

A sufficient condition for strict positivity of the variance process is the Feller condition $2\kappa\xi \geq \sigma^2$. When this condition is met, the boundary at zero is unattainable. If the condition fails, the process may reach zero, but the square-root diffusion remains non-negative under the standard CIR construction.

Unlike in the Black-Scholes model, the log-price is not normally distributed unconditionally because the variance path itself is random. We can see this when we apply Itô's lemma to $X_t = \log S_t$. Conditional on the variance path $\{v_s\}_{0 \leq s \leq t}$, the stochastic integral

$$X_t = X_0 + \int_0^t \left(r - \frac{1}{2} v_s \right) ds + \int_0^t \sqrt{v_s}\, dW_{S,s},$$

is Gaussian, but after integrating over all possible variance paths the distribution becomes more complex.

The correlation parameter ρ plays an important economic role. If $\rho < 0$, negative asset-price shocks tend to be associated with positive variance shocks. This produces a leverage effect and can generate implied-volatility skew. If $\rho > 0$, positive asset-price shocks tend to be associated with increasing variance. We normally see in equity markets, negative values of ρ are typically more common, although the appropriate value depends on the asset class and calibration data.

The Heston parameter vector is denoted $\boldsymbol{\theta}_{\mathrm{H}} = (v_0, \kappa, \xi, \sigma, \rho)$, with constraints

$$v_0 > 0, \qquad \kappa > 0, \qquad \xi > 0, \qquad \sigma > 0, \qquad \rho \in (-1, 1).$$

3 Analytical Benchmarks and Monte Carlo Pricing

The previous section defined the option payoffs and risk-neutral asset dynamics used throughout the paper. We now turn to the pricing methods associated with these models. The aim of this section is to identify a small number of mathematically reliable benchmark cases and then explain how Monte Carlo pricing is used when such closed-form expressions are unavailable or unsuitable.

The two analytical benchmark prices used in this paper are the Black-Scholes geometric Asian call and the Black-Scholes Margrabe exchange option. The geometric Asian option provides a tractable path-dependent benchmark because the logarithm of the geometric average is normally distributed under Black-Scholes dynamics. The Margrabe option provides a tractable multi-asset benchmark because, under a two-asset Black-Scholes model, the option can be reduced to a call option on the exchange ratio. These two formulas will be derived in full in this section.

For models such as Merton jump-diffusion and Heston stochastic volatility, exotic option prices will instead be computed by Monte Carlo simulation. This choice reflects the difficulty of deriving closed-form solutions for path-dependent claims under these models. For path-dependent options, Monte Carlo methods provide a flexible way to evaluate risk-neutral expectations by simulating sample paths and averaging discounted payoffs. The same pricing engine can then be embedded inside the Bayesian calibration framework introduced in the next section.

3.1 Asian Options Under Black-Scholes

Take the probability space $(\Omega, \mathcal{F}, \mathbb{Q})$, where $\mathcal{F}$ is the filtration and $\mathbb{Q}$ is the risk-neutral measure. Let A_T denote the payoff functional of a geometric Asian contract with maturity T. Under a risk-neutral measure $\mathbb{Q}$ and a constant risk-free rate r, the time-zero price is

$$C_0 = e^{-rT}\mathbb{E}_{\mathbb{Q}}[A_T \mid \mathcal{F}_0],$$

where,

$$A_T = \left(\exp\left[\frac{1}{T}\int_0^T \log S_t\, dt\right] - K\right)^+$$

We note here that for a path-dependent payoff, such as an Asian option, the expectation depends on the joint distribution of the asset path. Recall in the previous section that under the risk-neutral measure, the asset price satisfies

$$X_t = X_0 + \left(r - \frac{1}{2}\sigma^2\right)t + \sigma W_t, \tag{4}$$

where $X_0 = \log S_0$. Define the average log-price as $\bar{X}_T = \frac{1}{T}\int_0^T X_t\, dt$, then

$$A_T = \left(e^{\bar{X}_T} - K\right)^+$$

Substituting (4) into $\bar{X}_T$, we obtain

$$\begin{aligned}\bar{X}_T &= \frac{1}{T}\int_0^T \left[X_0 + \left(r - \frac{1}{2}\sigma^2\right)t + \sigma W_t\right] dt,\\ &= X_0 + \left(r - \frac{1}{2}\sigma^2\right)\frac{1}{T}\int_0^T t\,dt + \frac{\sigma}{T}\int_0^T W_t\,dt,\\ &= \log S_0 + \left(r - \frac{1}{2}\sigma^2\right)\frac{T}{2} + \frac{\sigma}{T}\int_0^T W_t\,dt.\end{aligned}$$

The random variable $\int_0^T W_t\,dt$ is Gaussian because it is a linear functional of a Gaussian process. Its mean is $\mathbb{E}_{\mathbb{Q}}\left[\int_0^T W_t\,dt\right] = \int_0^T \mathbb{E}_{\mathbb{Q}}[W_t]\,dt = 0$. Its variance is

$$\begin{aligned}\mathbb{V}\mathrm{ar}_{\mathbb{Q}}\left(\int_0^T W_t\,dt\right) &= \mathbb{E}_{\mathbb{Q}}\left[\left(\int_0^T W_t\,dt\right)^2\right],\\ &= \mathbb{E}_{\mathbb{Q}}\left[\int_0^T\int_0^T W_s W_t\,ds\,dt\right],\\ &= \int_0^T\int_0^T \mathbb{E}_{\mathbb{Q}}[W_s W_t]\,ds\,dt.\end{aligned}$$

For Brownian motion, $\mathbb{E}_{\mathbb{Q}}[W_s W_t] = \min(s,t)$. Therefore, if we split the square $[0,T]^2$ into the regions $s \le t$ and $s > t$, by symmetry,

$$\begin{aligned}\mathbb{V}\mathrm{ar}^{\mathbb{Q}}\left(\int_0^T W_t\,dt\right) &= \int_0^T\int_0^T \min(s,t)\,ds\,dt,\\ &= 2\int_0^T\int_0^t s\,ds\,dt,\\ &= 2\int_0^T \frac{t^2}{2}\,dt,\\ &= \int_0^T t^2\,dt,\\ &= \frac{T^3}{3}.\end{aligned}$$

Hence,

$$\int_0^T W_t\,dt \sim \mathcal{N}\left(0, \frac{T^3}{3}\right).$$

This then gives us the distribution for $\bar{X}_T$:

$$\bar{X}_T \sim \mathcal{N}(m_G, s_G^2), \tag{5}$$

where

$$m_G = \log S_0 + \left(r - \frac{1}{2}\sigma^2\right)\frac{T}{2}, \qquad s_G^2 = \frac{\sigma^2 T}{3}.$$

The pricing problem can now be reduced to the valuation of a call option on a lognormal random variable. Specifically,

$$C_0^A = e^{-rT}\mathbb{E}_{\mathbb{Q}}\left[(e^{\bar{X}_T} - K)^+ \mid \mathcal{F}_0\right], \tag{6}$$

with $\bar{X}_T$ defined above. Let

$$Y \sim \mathcal{N}(m, s^2), \qquad s > 0.$$

Then, for a real-valued probability measure $\mathbb{P}$,

$$\begin{aligned}\mathbb{E}\left[(e^Y - K)^+\right] &= \mathbb{E}\left[(e^Y - K)\mathbf{1}_{\{e^Y > K\}}\right], \\ &= \mathbb{E}\left[e^Y \mathbf{1}_{\{Y > \log K\}}\right] - K\mathbb{P}(Y > \log K).\end{aligned}$$

The second term is immediate:

$$\mathbb{P}(Y > \log K) = \Phi\left(\frac{m - \log K}{s}\right),$$

where Φ denotes the standard normal cumulative distribution function. For the first term,

$$\begin{aligned}\mathbb{E}\left[e^Y \mathbf{1}_{\{Y > \log K\}}\right] &= \int_{\log K}^{\infty} e^y \frac{1}{s\sqrt{2\pi}} \exp\left[-\frac{(y-m)^2}{2s^2}\right] dy \\ &= \int_{\log K}^{\infty} \frac{1}{s\sqrt{2\pi}} \exp\left[y - \frac{(y-m)^2}{2s^2}\right] dy,\end{aligned}$$

Note that $y^2 - 2(m+s^2)y + m^2 = (y - (m+s^2))^2 - (m+s^2)^2 + m^2$, and $y - \frac{(y-m)^2}{2s^2} = -\frac{(y-(m+s^2))^2}{2s^2} + \frac{(m+s^2)^2 - m^2}{2s^2} = -\frac{(y-(m+s^2))^2}{2s^2} + m + \frac{1}{2}s^2$, thus

$$\begin{aligned}\mathbb{E}\left[e^Y \mathbf{1}_{\{Y > \log K\}}\right] &= e^{m+\frac{1}{2}s^2} \int_{\log K}^{\infty} \frac{1}{s\sqrt{2\pi}} \exp\left[-\frac{(y-(m+s^2))^2}{2s^2}\right] dy \\ &= e^{m+\frac{1}{2}s^2} \Phi\left(\frac{m + s^2 - \log K}{s}\right).\end{aligned}$$

Combining the two terms, we obtain

$$\mathbb{E}\left[(e^Y - K)^+\right] = e^{m+\frac{1}{2}s^2}\Phi(d_1) - K\Phi(d_2), \tag{7}$$

where

$$d_1 = \frac{m - \log K + s^2}{s}, \qquad d_2 = \frac{m - \log K}{s} = d_1 - s.$$

Now we apply (7) with $Y = \bar{X}_T$, $m = m_G$, and $s = s_G$, giving us the Black-Scholes price of the continuously monitored geometric Asian call is

$$C_0^A = e^{-rT}\left[e^{m_G + \frac{1}{2}s_G^2}\Phi(d_1^G) - K\Phi(d_2^G)\right], \tag{8}$$

where

$$d_1^G = \frac{m_G - \log K + s_G^2}{s_G}, \qquad d_2^G = d_1^G - s_G.$$

3.2 Margrabe Options Under Black-Scholes

For the Margrabe exchange option, let $S_{1,t}$ and $S_{2,t}$ be two risky assets satisfying

$$\frac{dS_{1,t}}{S_{1,t}} = r\,dt + \sigma_1\,dW_{1,t}, \qquad \frac{dS_{2,t}}{S_{2,t}} = r\,dt + \sigma_2\,dW_{2,t},$$

where

$$d\langle W_1, W_2\rangle_t = \rho\, dt, \qquad \rho \in (-1, 1).$$

The price for its payoff at $t = 0$ is

$$C_0^M = e^{-rT}\mathbb{E}_{\mathbb{Q}}\left[(S_{1,T} - S_{2,T})^+ \mid \mathcal{F}_0\right]. \tag{9}$$

Although this payoff is not path-dependent, it is multi-asset. Its value depends on the joint distribution of $S_{1,T}$ and $S_{2,T}$, and in particular on the correlation between the two assets. We can rewrite the payoff in terms of the exchange ratio $R_t = \frac{S_{1,t}}{S_{2,t}}$, which means $(S_{1,T} - S_{2,T})^+ = S_{2,T}(R_T - 1)^+$.

To evaluate this expectation, use $S_{2,t}$ as numeraire. Define a new probability measure $\mathbb{Q}^{(2)}$ on $(\mathcal{F}_t)_{t\in[0,T]}$ by

$$\left.\frac{d\mathbb{Q}^{(2)}}{d\mathbb{Q}}\right|_{\mathcal{F}_T} = \frac{e^{-rT}S_{2,T}}{S_{2,0}}. \tag{10}$$

Since $e^{-rt}S_{2,t}$ is a martingale under $\mathbb{Q}$, the Radon–Nikodym derivative in (10) has expectation one and therefore defines an equivalent probability measure. Using (10), equation (9) becomes

$$C_0^M = S_{2,0}\mathbb{E}_{\mathbb{Q}^{(2)}}\left[(R_T - 1)^+ \mid \mathcal{F}_0\right].$$

We now need to derive the dynamics of R_t under $\mathbb{Q}^{(2)}$, and since

$$\log R_t = \log S_{1,t} - \log S_{2,t}, \tag{11}$$

we have

$$\begin{aligned} d\log R_t &= \left(r - \frac{1}{2}\sigma_1^2\right)dt + \sigma_1\, dW_{1,t} - \left(r - \frac{1}{2}\sigma_2^2\right)dt - \sigma_2\, dW_{2,t} \\ &= -\frac{1}{2}(\sigma_1^2 - \sigma_2^2)dt + \sigma_1\, dW_{1,t} - \sigma_2\, dW_{2,t}. \end{aligned}$$

Recall, $\sigma_{\text{ex}} = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$. Under the S_2-numeraire measure $\mathbb{Q}^{(2)}$, the exchange ratio $R_t = S_{1,t}/S_{2,t}$ is a tradable asset expressed in units of the numeraire. Therefore, it is a martingale under $\mathbb{Q}^{(2)}$. Its dynamics are

$$\frac{dR_t}{R_t} = \sigma_{\text{ex}}\, dW_t^{(2)}, \tag{12}$$

where $W_t^{(2)}$ is a Brownian motion under $\mathbb{Q}^{(2)}$. Applying Itô's lemma to $\log R_t$ under (12), we obtain

$$d\log R_t = -\frac{1}{2}\sigma_{\text{ex}}^2\, dt + \sigma_{\text{ex}}\, dW_t^{(2)}.$$

Therefore,

$$\log R_T = \log R_0 - \frac{1}{2}\sigma_{\text{ex}}^2 T + \sigma_{\text{ex}} W_T^{(2)}.$$

Thus

$$\log R_T \sim \mathcal{N}\left(\log R_0 - \frac{1}{2}\sigma_{\text{ex}}^2 T, \sigma_{\text{ex}}^2 T\right) \quad \text{under } \mathbb{Q}^{(2)}.$$

Let $Y = \log R_T \sim \mathcal{N}(m_R, s_R^2)$, then by applying the lognormal call identity to $R_T = e^Y$, where

$$m_R = \log R_0 - \frac{1}{2}\sigma_{\text{ex}}^2 T, \qquad s_R^2 = \sigma_{\text{ex}}^2 T,$$

to

$$\mathbb{E}\left[(e^Y - K)^+\right] = \mathbb{E}\left[e^Y \mathbf{1}_{\{Y>\log K\}}\right] - K\mathbb{P}(Y > \log K) = e^{m+\frac{1}{2}s^2}\Phi(d_1) - K\Phi(d_2),$$

where $K = 1$, gives

$$\mathbb{E}_{\mathbb{Q}^{(2)}}\left[(R_T - 1)^+\right] = e^{m_R+\frac{1}{2}s_R^2}\Phi(d_1^M) - \Phi(d_2^M).$$

Since $e^{m_R+\frac{1}{2}s_R^2} = e^{\log R_0} = R_0$, we obtain the final closed-form

$$\begin{aligned} C_0^M &= S_{2,0}\left[R_0\Phi(d_1^M) - \Phi(d_2^M)\right], \\ &= S_{1,0}\Phi(d_1^M) - S_{2,0}\Phi(d_2^M), \end{aligned}$$

where

$$d_1^M = \frac{\log(S_{1,0}/S_{2,0}) + \frac{1}{2}\sigma_{\text{ex}}^2 T}{\sigma_{\text{ex}}\sqrt{T}}, \qquad d_2^M = d_1^M - \sigma_{\text{ex}}\sqrt{T}.$$

The effective volatility clarifies the role of correlation. If the two assets have the same volatility and are perfectly positively correlated, then $\sigma_{\text{ex}} = 0$, and the exchange ratio has no randomness. Otherwise the option has time value.

3.3 Pricing for Merton and Heston Exotics

The analytical formulas above are useful because they are exact under Black-Scholes assumptions. However, the main models of interest in this paper include jumps and stochastic volatility. In such models, exotic option prices generally do not reduce to simple lognormal expectations. Monte Carlo pricing is therefore used to approximate risk-neutral expectations directly. Let

$$C(\boldsymbol{\theta}) = e^{-rT}\mathbb{E}_{\boldsymbol{\theta}}^{\mathbb{Q}}[g]$$

be the model-implied price of a derivative with payoff g. Suppose that we generate M independent sample paths under the model with parameter vector $\boldsymbol{\theta}$. Let $g^{(m)}$ denote the payoff evaluated on the m-th simulated path. The standard Monte Carlo estimator is

$$\widehat{C}_M(\boldsymbol{\theta}) = e^{-rT}\frac{1}{M}\sum_{m=1}^{M} g^{(m)}.$$

If the simulated payoffs are independent and identically distributed with finite variance, then

$$\mathbb{E}\left[\widehat{C}_M(\boldsymbol{\theta})\right] = C(\boldsymbol{\theta}), \tag{13}$$

provided that the simulated paths are exact samples from the model and the payoff is evaluated exactly. In practice, when the asset path is discretised, the estimator may also contain time-discretisation bias. The variance of the Monte Carlo estimator is

$$\mathbb{V}\text{ar}\left[\widehat{C}_M(\boldsymbol{\theta})\right] = e^{-2rT}\frac{1}{M}\,\mathbb{V}\text{ar}(g), \tag{14}$$

where the variance is taken under the risk-neutral model. Therefore the standard error is

$$\text{SE}\left[\widehat{C}_M(\boldsymbol{\theta})\right] = e^{-rT}\frac{\sqrt{\mathbb{V}\text{ar}(g)}}{\sqrt{M}}.$$

Since $\mathbb{V}\mathrm{ar}(g)$ is usually unknown, it is estimated by the sample variance

$$\widehat{\mathbb{V}\mathrm{ar}}(g) = \frac{1}{M-1}\sum_{m=1}^{M}\left(g^{(m)} - \bar{g}_M\right)^2,$$

where $\bar{g}_M = \frac{1}{M}\sum_{m=1}^{M} g^{(m)}$. The estimated standard error is then

$$\widehat{\mathrm{SE}}\left[\widehat{C}_M(\boldsymbol{\theta})\right] = e^{-rT}\frac{\sqrt{\widehat{\mathbb{V}\mathrm{ar}}(g)}}{\sqrt{M}}.$$

By the central limit theorem,

$$\sqrt{M}\left(\widehat{C}_M(\boldsymbol{\theta}) - C(\boldsymbol{\theta})\right) \to \mathcal{N}\left(0, e^{-2rT}\,\mathbb{V}\mathrm{ar}(g)\right), \qquad M \to \infty.$$

This implies that the convergence rate of standard Monte Carlo is $M^{-1/2}$, independent of the dimension of the underlying path.

Many exotic payoffs depend on the full path of the underlying asset. Since numerical simulation can only generate a finite number of time points, the continuous-time path must be approximated on a time grid. Let $0 = t_0 < t_1 < \cdots < t_N = T$ be a uniform time grid with $\Delta t = \frac{T}{N}$. For a simulated path, write $S_j = S_{t_j}, \quad j = 0, \ldots, N$. Continuously monitored geometric Asian option must be approximated. A simple grid approximation is

$$A_N = \exp\left(\frac{1}{N}\sum_{j=1}^{N}\log S_j\right),$$

for right-point monitoring, or

$$A_N^{\mathrm{geom,trap}} = \exp\left(\frac{1}{2N}\left[\log S_0 + 2\sum_{j=1}^{N-1}\log S_j + \log S_N\right]\right)$$

for trapezoidal approximation.

For the Margrabe exchange option, no path approximation is required under the standard terminal payoff, since the payoff only depends on terminal values.

4 Markov Chain Monte Carlo Techniques

The previous section described how model-implied exotic option prices may be computed either analytically, in benchmark cases, or numerically by Monte Carlo simulation. We now turn to the calibration problem. The objective of calibration is to infer unknown model parameters from a collection of option prices. Classical calibration usually proceeds by choosing the parameter vector that minimises a pricing error criterion. If y_i denotes the observed price of the i-th option and $C_i(\boldsymbol{\theta})$ denotes the model-implied price under parameter vector $\boldsymbol{\theta}$, then a standard least-squares calibration solves

$$\widehat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} \sum_{i=1}^{n} \left(y_i - C_i(\boldsymbol{\theta})\right)^2 .$$

This approach is simple and widely used, but it produces only a point estimate. It does not directly describe uncertainty in the calibrated parameters, nor does it quantify how this uncertainty propagates into exotic option prices. In contrast, a Bayesian approach treats the model parameters as unknown random variables. Bayesian calibration produces a posterior distribution over plausible parameter values conditional on the observed option price data. This is particularly useful in exotic option pricing because the calibration problem is often nonlinear, high-dimensional, and weakly identified. Several different parameter combinations may produce similar option prices, especially when the observed contracts cover only a limited range of strikes or maturities. A posterior distribution can reveal this uncertainty, whereas a point estimate may conceal it.

The purpose of this section is to formulate the Bayesian calibration problem and introduce the Markov chain Monte Carlo methods used to sample from the resulting posterior distributions. We first define the likelihood and posterior distribution. We then discuss effective sample size, which is used to evaluate the quality of MCMC output. After this, we introduce the Metropolis-Hastings algorithm, adaptive Metropolis-Hastings, Metropolis-Hastings within Gibbs, birth-death MCMC, and Hamiltonian Monte Carlo.

4.0.1 Bayesian Calibration

Suppose that we observe a collection of n option prices $\boldsymbol{y} = (y_1, \ldots, y_n)^\top$. Each observation y_i corresponds to a contract with known characteristics, such as strike K_i, maturity T_i, payoff type, and underlying asset information. Let $\mathcal{I}_i$ denote the full set of contract information for the i-th option. For example, for an Asian option this may include $\mathcal{I}_i = (S_0, K_i, T_i, r, \text{monitoring convention})$, whereas for a Margrabe option it may include $\mathcal{I}_i = (S_{1,0}, S_{2,0}, T_i, r)$. Let $\boldsymbol{\theta}$ denote the parameter vector of the pricing model. The model-implied price of the i-th contract is written as

$$C_i(\boldsymbol{\theta}) = C(\boldsymbol{\theta}; \mathcal{I}_i) = e^{-rT_i} \mathbb{E}_{\mathbb{Q},\, \boldsymbol{\theta}} \left[g_i\right],$$

A natural observation model is

$$y_i = C_i(\boldsymbol{\theta}) + \varepsilon_i, \qquad i = 1, \ldots, n,$$

where $\varepsilon_i \sim \mathcal{N}(0, \tau_i^2)$ represents pricing noise and is independent across contracts. Under this assumption,

$$y_i \mid \boldsymbol{\theta}, \tau_i \sim \mathcal{N}\left(C_i(\boldsymbol{\theta}), \tau_i^2\right).$$

The likelihood is therefore

$$p(\boldsymbol{y} \mid \boldsymbol{\theta}, \boldsymbol{\tau}) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\tau_i^2}} \exp\left[-\frac{(y_i - C_i(\boldsymbol{\theta}))^2}{2\tau_i^2}\right],$$

where $\boldsymbol{\tau} = (\tau_1, \ldots, \tau_n)^\top$. However, it is convenient to assume a common observation noise level $\tau_i = \tau$. Taking logs gives the log-likelihood

$$\ell(\boldsymbol{\theta}, \tau; \boldsymbol{y}) = -\frac{n}{2}\log(2\pi\tau^2) - \frac{1}{2\tau^2}\sum_{i=1}^{n}(y_i - C_i(\boldsymbol{\theta}))^2.$$

The Bayesian posterior distribution follows from Bayes' theorem. Let $\pi(\boldsymbol{\theta})$ denote the prior distribution on the model parameters and let $\pi(\boldsymbol{\tau})$ denote the prior distribution on the observation noise parameters. Then

$$\pi(\boldsymbol{\theta}, \boldsymbol{\tau} \mid \boldsymbol{y}) = \frac{p(\boldsymbol{y} \mid \boldsymbol{\theta}, \boldsymbol{\tau})\pi(\boldsymbol{\theta})\pi(\boldsymbol{\tau})}{p(\boldsymbol{y})}, \tag{15}$$

where

$$p(\boldsymbol{y}) = \int p(\boldsymbol{y} \mid \boldsymbol{\theta}, \boldsymbol{\tau})\pi(\boldsymbol{\theta})\pi(\boldsymbol{\tau})\, d\boldsymbol{\theta}\, d\boldsymbol{\tau}$$

is the marginal likelihood. Since $p(\boldsymbol{y})$ does not depend on the unknown parameters, MCMC algorithms only require the posterior density up to proportionality:

$$\pi(\boldsymbol{\theta}, \boldsymbol{\tau} \mid \boldsymbol{y}) \propto p(\boldsymbol{y} \mid \boldsymbol{\theta}, \boldsymbol{\tau})\pi(\boldsymbol{\theta})\pi(\boldsymbol{\tau}).$$

Equivalently, the log-posterior is

$$\log \pi(\boldsymbol{\theta}, \boldsymbol{\tau} \mid \boldsymbol{y}) = \ell(\boldsymbol{\theta}, \boldsymbol{\tau}; \boldsymbol{y}) + \log \pi(\boldsymbol{\theta}) + \log \pi(\boldsymbol{\tau}) + c,$$

where c is a constant independent of $\boldsymbol{\theta}$ and $\boldsymbol{\tau}$.

The parameters in option pricing models are constrained. For instance, volatility parameters must be positive, jump intensities must be non-negative, variance parameters must be positive, and correlation parameters must lie in $(-1, 1)$. It is therefore convenient for us to transform the constrained parameter vector into an unconstrained vector. Let $\boldsymbol{\eta} \in \mathbb{R}^d$ denote the unconstrained parameter vector, and let $\boldsymbol{\theta} = T(\boldsymbol{\eta})$ be a differentiable transformation from unconstrained space into the valid parameter space. For a positive parameter $a > 0$, we use the exponential transformation

$$a = \exp(\eta_a), \qquad \eta_a \in \mathbb{R}.$$

For a correlation parameter $\rho \in (-1, 1)$, we use

$$\rho = \tanh(\eta_\rho), \qquad \eta_\rho \in \mathbb{R}.$$

The inverse transformation is

$$\eta_\rho = \operatorname{arctanh}(\rho) = \frac{1}{2}\log\left(\frac{1+\rho}{1-\rho}\right).$$

If the prior is specified on the constrained parameter vector $\boldsymbol{\theta}$, then the posterior density on the unconstrained vector $\boldsymbol{\eta}$ must include the Jacobian of the transformation:

$$\pi_\eta(\boldsymbol{\eta} \mid \boldsymbol{y}) = \pi_\theta(T(\boldsymbol{\eta}) \mid \boldsymbol{y}) \left|\det J_T(\boldsymbol{\eta})\right|, \tag{16}$$

where $J_T(\boldsymbol{\eta})$ is the Jacobian matrix of T. Hence

$$\log \pi_\eta(\boldsymbol{\eta} \mid \boldsymbol{y}) = \log \pi_\theta(T(\boldsymbol{\eta}) \mid \boldsymbol{y}) + \log \left|\det J_T(\boldsymbol{\eta})\right|. \tag{17}$$

If instead the prior is specified directly on $\boldsymbol{\eta}$, then no additional Jacobian term is required.

The posterior distribution usually has no closed-form normalising constant and cannot be sampled from directly. Hence, the role of MCMC is to construct a sequence of dependent samples $\boldsymbol{\eta}^{(1)}, \boldsymbol{\eta}^{(2)}, \ldots, \boldsymbol{\eta}^{(K)}$ whose stationary distribution is the posterior distribution $\pi_\eta(\boldsymbol{\eta} \mid \boldsymbol{y})$. Once posterior samples have been obtained, they can be transformed back to the constrained parameter scale:

$$\boldsymbol{\theta}^{(k)} = T(\boldsymbol{\eta}^{(k)}), \qquad k = 1, \ldots, K. \tag{18}$$

These samples can then be used to estimate posterior means, credible intervals, posterior predictive prices, and uncertainty in exotic option values.

4.0.2 Effective Sample Size

MCMC samples are not independent. Each draw in a Markov chain depends on the previous state, and this serial dependence reduces the amount of independent information contained in the sample. *Effective sample size (ESS)* measures how many independent samples would contain approximately the same amount of information as the correlated MCMC sample. Let $h^{(k)} = h(\boldsymbol{\eta}^{(k)})$ be a scalar function of the k-th MCMC draw. For example, h may select one parameter component, such as a volatility or correlation parameter. If the samples were drawn independently and identically from π, then we can define the *lag-ℓ autocorrelation* of the chain as

$$\rho_\ell = \frac{\mathbb{C}\mathrm{ov}(h^{(k)}, h^{(k+\ell)})}{\mathbb{V}\mathrm{ar}(h^{(k)})}.$$

Under standard regularity conditions, the variance of the MCMC sample mean is approximately

$$\mathbb{V}\mathrm{ar}(\bar{h}_K) \approx \frac{\mathbb{V}\mathrm{ar}(h)}{K} \left(1 + 2 \sum_{\ell=1}^{\infty} \rho_\ell\right).$$

This motivates the definition

$$\mathrm{ESS} = \frac{K}{1 + 2 \sum_{\ell=1}^{\infty} \rho_\ell}. \tag{19}$$

If the samples are independent, then $\rho_\ell = 0, \ \forall\ \ell \geq 1$, and therefore $\mathrm{ESS} = K$.

Typically, we examine the ESS for each component of $\boldsymbol{\theta}$ to assess whether sufficient mixing has occurred. A sufficient ESS depends on the desired accuracy. In practice, an ESS of about 500 is often considered adequate, whereas an ESS below 100 suggests that inference from the MCMC sample should be treated with caution due to poor mixing.

4.1 Metropolis-Hastings

The Metropolis-Hastings algorithm constructs a Markov chain whose stationary distribution is the posterior distribution. Let $\widetilde{\pi}(\boldsymbol{\eta}) \propto \pi(\boldsymbol{\eta} \mid \boldsymbol{y})$ denote the unnormalised posterior density. At iteration k, suppose the current state is $\boldsymbol{\eta}^{(k)}$. A proposal density

$$q(\boldsymbol{\eta}' \mid \boldsymbol{\eta}^{(k)})$$

is used to generate a candidate value $\boldsymbol{\eta}'$. The proposed move is accepted with probability

$$\alpha\left(\boldsymbol{\eta}^{(k)}, \boldsymbol{\eta}'\right) = \min\left\{1, \frac{\widetilde{\pi}(\boldsymbol{\eta}')q(\boldsymbol{\eta}^{(k)} \mid \boldsymbol{\eta}')}{\widetilde{\pi}(\boldsymbol{\eta}^{(k)})q(\boldsymbol{\eta}' \mid \boldsymbol{\eta}^{(k)})}\right\}. \tag{20}$$

The proposal-density ratio corrects for asymmetry in the proposal. If the proposal is symmetric, then

$$q(\boldsymbol{\eta}' \mid \boldsymbol{\eta}^{(k)}) = q(\boldsymbol{\eta}^{(k)} \mid \boldsymbol{\eta}'),$$

and the acceptance probability becomes

$$\alpha = \min\left\{1, \frac{\widetilde{\pi}(\boldsymbol{\eta}')}{\widetilde{\pi}(\boldsymbol{\eta}^{(k)})}\right\}.$$

In computation, the acceptance probability is evaluated on the log scale:

$$\log\alpha = \min\Bigg\{0, \log\widetilde{\pi}(\boldsymbol{\eta}') - \log\widetilde{\pi}(\boldsymbol{\eta}^{(k)}) \\ + \log q(\boldsymbol{\eta}^{(k)} \mid \boldsymbol{\eta}') - \log q(\boldsymbol{\eta}' \mid \boldsymbol{\eta}^{(k)})\Bigg\}.$$

Then draw $u \sim \text{Uniform}(0,1)$ and set

$$\boldsymbol{\eta}^{(k+1)} = \begin{cases} \boldsymbol{\eta}', & \text{if } \log u < \log\alpha, \\ \boldsymbol{\eta}^{(k)}, & \text{otherwise.} \end{cases}$$

The *random-walk Metropolis algorithm* then uses

$$\boldsymbol{\eta}' = \boldsymbol{\eta}^{(k)} + \boldsymbol{\epsilon}^{(k)}, \qquad \boldsymbol{\epsilon}^{(k)} \sim \mathcal{N}(\mathbf{0}, \Sigma_q).$$

The proposal covariance Σ_q controls the scale and direction of the proposed moves. If it is too small, the chain accepts many proposals but moves slowly. If it is too large, most proposals are rejected.

The empirical acceptance rate is

$$\text{AR} = \frac{1}{K}\sum_{k=1}^{K} \mathbf{1}\left\{\text{proposal at iteration } k \text{ was accepted}\right\}.$$

It is important to note that in practical cases, we aim to have the acceptance rate around 0.2-0.4.

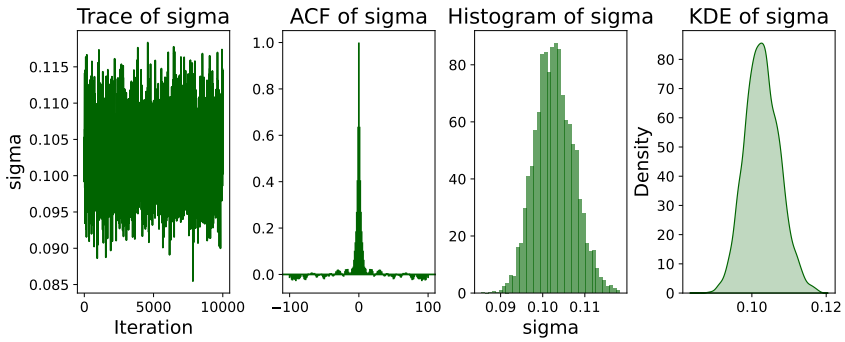


Figure 1: Diagnostic plots for the random walk Metropolis sampler for σ: (i) trace plots, (ii) autocorrelation functions (ACF), (iii) histograms, and (iv) kernel density estimates (KDE). The trace shows stationary exploration with the ACF decaying rapidly to near zero. The distribution are unimodal and approximately Gaussian.

Figure 1 shows the geometry of the Black-Scholes likelihood and the local nature of a random walk proposal. Under discretely observed log-returns, the log likelihood for σ is close to quadratic in a neighbourhood of its maximiser, so the posterior for σ is well approximated by a normal law on the working scale. This is visible in the distribution plots, which are unimodal and nearly symmetric with small variance, and in the trace, which remains confined to a narrow band about 0.103 while exhibiting frequent level crossings after burn-in. The ACF shows a positive short-lag correlation that dissipates over moderate lags, which yields an ESS of roughly 2.1×10^3 in Table **??**. This shows a well-tuned one-dimensional random walk that nonetheless explores by local diffusion rather than by long, directed moves.

4.1.1 Adaptive Metropolis-Hastings

Random-walk Metropolis can be inefficient when the posterior has strong correlations or very different scales across parameters. Adaptive Metropolis-Hastings improves the proposal by estimating the posterior covariance from the chain during a warm-up period.

A standard random walk metropolis algorithm uses a fixed proposal covariance, C. If C is too small, the chain mixes slowly (high autocorrelation). On the other hand, if too large, proposals are often rejected. In high or moderate dimensions, optimal choices of C depend on the correlation structure of the target posterior.

We want to let the chain learn an approximately optimal proposal shape at each iteration by estimating the posterior's local covariance structure from past samples. The ideal proposal for a Gaussian target $\mathcal{N}(\mu, \Sigma)$ is $\mathcal{N}(\theta, \Sigma)$. *Adaptive Metropolis-Hastings (AMH)* seeks to mimic this by continually updating an empirical covariance, allowing the proposal distribution to evolve in response to information gathered from past samples and thereby automatically tune itself to the shape of the target posterior.

Say we had a two risky assets S_1, S_2 under the Black-Scholes, with correlation $\rho \in (-1, 1)$, such that:

$$\begin{aligned} dS_{1,t} &= rS_{1,t}dt + \sigma S_{1,t}dW^{\mathbb{Q}}_{1,t}, \\ dS_{2,t} &= rS_{2,t}dt + \nu S_{2,t}dW^{\mathbb{Q}}_{2,t}, \\ d\langle W^{\mathbb{Q}}_{1,t}, W^{\mathbb{Q}}_{2,t}\rangle &= \rho dt, \end{aligned}$$

where $\sigma, \nu > 0$, and $W_{i,t}^{\mathbb{Q}}$ are the Wiener processes under the risk-neutral measure for each risky asset $i = 1, 2$. Let $\boldsymbol{\theta} \in \mathbb{R}^3$ denote the current state of the Markov chain, and let $\pi(\boldsymbol{\theta}|\boldsymbol{X})$ be the three-dimensional posterior over (σ, ν, ρ) for the two-asset Black-Scholes model, reparameterised via $\theta_1 = \ln \sigma$, $\theta_2 = \ln \nu$, and $\theta_3 = \operatorname{arctanh}(\rho)$.

At iteration n, given the current state $\boldsymbol{\theta}^{(n)}$ and a symmetric positive-definite proposal covariance $C^{(n)}$, the AMH algorithm generates a candidate $\boldsymbol{\theta}'$ by drawing from the Gaussian proposal distribution $\mathcal{N}(\boldsymbol{\theta}^{(n)}, C^{(n)})$. The acceptance probability for the Metropolis algorithm is then computed as

$$\alpha(\boldsymbol{\theta}^{(n)}, \boldsymbol{\theta}') = \min\left(1, \frac{\pi(\boldsymbol{\theta}'|\boldsymbol{X}) q(\boldsymbol{\theta}^{(n)}|\boldsymbol{\theta}')}{\pi(\boldsymbol{\theta}^{(n)}|\boldsymbol{X}) q(\boldsymbol{\theta}'|\boldsymbol{\theta}^{(n)})}\right) = \min\left(1, \frac{\pi(\boldsymbol{\theta}'|\boldsymbol{X})}{\pi(\boldsymbol{\theta}^{(n)}|\boldsymbol{X})}\right),$$

and $\boldsymbol{\theta}^{(n+1)}$ is set equal to $\boldsymbol{\theta}'$ with probability α, or remains at $\boldsymbol{\theta}^{(n)}$ otherwise. Upon accepting or rejecting this proposal, we update the empirical mean $M^{(n)}$ and sum of squares matrix $S^{(n)}$ using the *one-pass algorithms* (Welford, 1962):

$$\begin{aligned} M^{(n+1)} &= M^{(n)} + \frac{\boldsymbol{\theta}^{(n+1)} - M^{(n)}}{n+1} \\ S^{(n+1)} &= S^{(n)} + (\boldsymbol{\theta}^{(n+1)} - M^{(n)})(\boldsymbol{\theta}^{(n+1)} - M^{(n+1)})^{\top}, \end{aligned}$$

so that the empirical covariance is $\hat{\Sigma}^{(n+1)} = \frac{S^{(n+1)}}{n}$ $\forall n \geq 1$. The idea of adaptation consists in blending the previous proposal covariance $C^{(n)}$ with a scaled version of the empirical covariance. Specifically, after a predetermined burn-in period of n_0 iterations, we define the scaled empirical covariance as

$$\Sigma_{\text{emp}}^{(n+1)} = \frac{(2.38)^2}{d} \hat{\Sigma}^{(n+1)} + \epsilon I_d,$$

where the factor $\frac{(2.38)^2}{d}$ is known from the optimal-scaling theory of Gaussian targets to make an asymptotic acceptance rate of approximately 23.8%, and $\epsilon > 0$ is a small ridge term to ensure eigenvalues remain bounded away from zero (Gelman et al., 1996). Hence, the new proposal covariance is then set to

$$C^{(n+1)} = (1 - \gamma_n) C^{(n)} + \gamma_n \Sigma_{\text{emp}}^{(n+1)}, \qquad \gamma_n \to 0,$$

with an adaptation schedule whose update size decreases with n. For $n < n_0$, we let $C^{(n+1)} = C^{(n)}$. The two theoretical properties that guarantee the AMH chain remains ergodic with the correct stationary distribution are the *diminishing adaptation condition* and the *containment condition*. The former condition requires that the total-variation distance between successive proposal kernels tends to zero. In other words, if $K_n(\boldsymbol{\theta}, \cdot)$ is the proposal kernel at step n, then we require

$$\sup_{\boldsymbol{\theta}} ||K_{n+1}(\boldsymbol{\theta}, \cdot) - K_n(\boldsymbol{\theta}, \cdot)||_{\text{TV}} \longrightarrow 0 \quad \text{as } n \to \infty.$$

This is encouraged by using a decreasing adaptation weight $\gamma_n \to 0$ together with convergence of the empirical covariance estimate (Haario et al., 2001). The containment condition requires that the family of proposal kernels does not wander into explosive patterns. This is enforced by delaying adaptation until after n_0 iterations and by the ridge term ϵI_d, which keeps all eigenvalues within a controlled range.

Summary Statistics

Parameter	Mean	Standard Deviation	Effective Sample Size (ESS)
σ	-1.1344	0.0468	1 134.3722
ν	-1.3671	0.0448	755.0555
ρ	0.6548	0.0630	706.1350

Table 1: Posterior summary for the Adaptive Metropolis sampler applied to $\boldsymbol{\theta} = (\sigma, \nu, \rho)$. The table shows posterior means, posterior standard deviations, and univariate ESS. The large ESS values show the rapid mixing under covariance adaptation and an empirical acceptance rate of 0.2173.

The plots in Figure 2 show the behaviour expected from an adaptive Metropolis scheme with online covariance learning. After the burn-in at n_0, the mechanism rescales step sizes to the local posterior variance, rotates proposals along high-variance directions, and regularises small eigenmodes through ϵI_d. Consequently, the trace plots for σ, ν, ρ are stable and well-mixed after n_0, with frequent level crossings and no visible drift. Additionally, the low-lag ACF is materially reduced and decays quickly, which is reflected in the ESS figures in Table 1. The distributions are approximately Gaussian on the working scale for σ and ν, while ρ concentrates away from ± 1, avoiding boundary-induced autocorrelation. Any residual autocorrelation observed is attributable to bounded support and to finite-sample noise in $\hat{\Sigma}^{(n)}$ early in adaptation. Hence both effects diminish as n grows, which matches the visibly improved mixing deeper into the run.

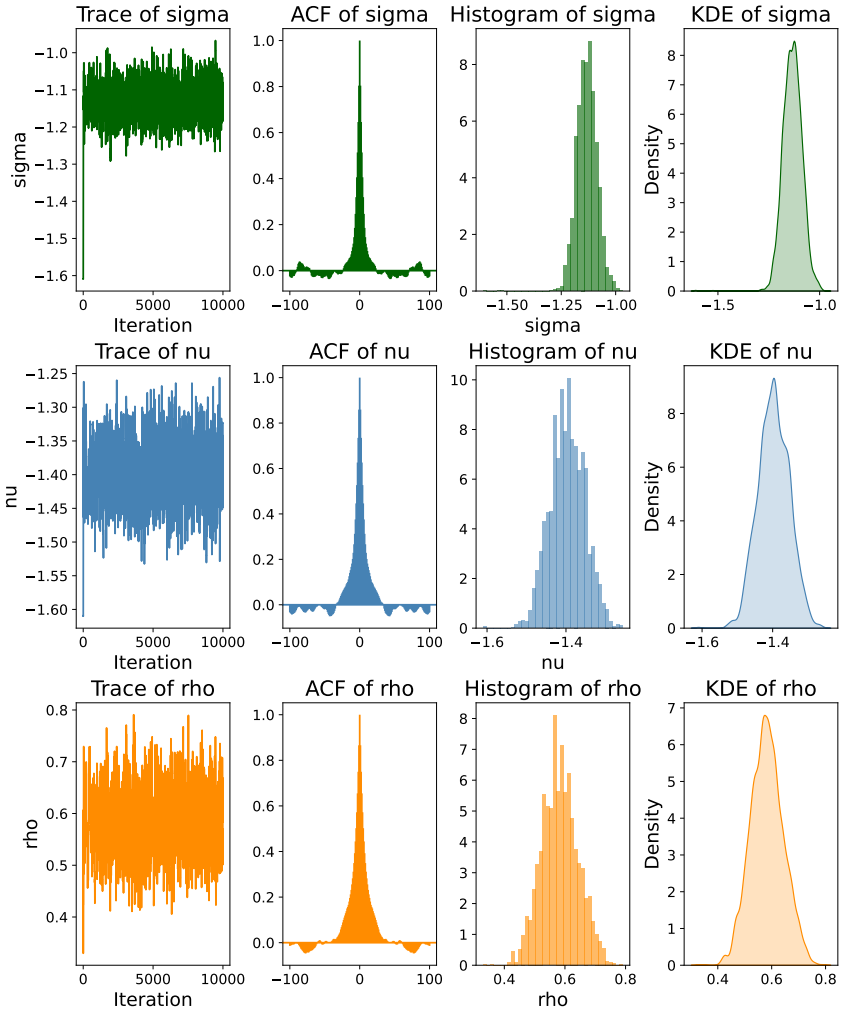


Figure 2: Adaptive MCMC diagnostic plots for each coordinate of $\boldsymbol{\theta} = (\sigma, \nu, \rho)$: (i) trace plots, (ii) autocorrelation functions (ACF), (iii) histograms, and (iv) kernel density estimates (KDE). The traces exhibit stationary exploration after burn-in, and the ACFs decay sharply relative to non-adaptive baselines. The distributions are unimodal with moderate spread, and ρ remains within its bounds with mass centred away from the boundaries.

4.2 Metropolis-Hastings within Gibbs

When applying MCMC to models with many parameters and latent variables, we apply *Gibbs sampling*. Gibbs sampling iteratively samples each parameter from its conditional distribution given all other variables. As such, we iteratively update latent states in blocks based on these distributions. However, when we do not have a closed-form distribution, we apply Metropolis-Hastings for that block. This is known as *Metropolis-within-Gibbs*. The idea is that we cycle through components of $\boldsymbol{\theta} = (\theta_1, \theta_2, \ldots, \theta_p)$, and for each θ_i we perform an MH update targeting the conditional density $\pi(\theta_i \mid \theta_{-i})$, for $i = 1, \ldots, p$, which leaves the full joint posterior $\pi(\theta_1, \ldots, \theta_p)$ invariant. We use Gibbs for the rest of our pricing models due to the high dimensionality of both the options and the complexity of the models.

The choice of block is flexible: we can update one parameter at a time, or a group of highly correlated parameters together. For example, in the Heston model, we can use a

Gibbs strategy that alternates between sampling the volatility latent path $v_{0:T}$ given the parameters, and sampling the parameters $(\kappa, \xi, \sigma, \rho)$ given the volatility path and price data. Consider the Heston model (without volatility jumps) below:

$$\begin{aligned} dS_t &= \mu S_t dt + S_t \sqrt{v_t} dW_{t,1}^{\mathbb{P}}, \\ dv_t &= \kappa(\xi - v_t)dt + \sigma\sqrt{v_t} dW_{t,2}^{\mathbb{P}}, \\ d\langle W_{t,1}^{\mathbb{P}}, W_{t,2}^{\mathbb{P}} \rangle &= \rho dt, \end{aligned}$$

where $2\kappa\xi > \sigma^2$, $\kappa, \sigma, \xi, \sqrt{v_t} > 0$, and $\rho \in (-1, 1)$. We can take $\boldsymbol{\theta} = (\kappa, \sigma, \xi, \rho)$. For pricing and calibration in this paper, we specify the Heston dynamics directly under the risk-neutral measure:

$$\begin{aligned} dS_t &= rS_t dt + \sqrt{v_t} S_t dW_t^{(1)}, \\ dv_t &= \kappa(\xi - v_t)dt + \sigma\sqrt{v_t} dW_t^{(2)}, \\ d\langle W_t^{(1)}, W_t^{(2)} \rangle &= \rho dt, \end{aligned}$$

where $r > 0$ is the risk-free rate, and $W_t^{(1)}, W_t^{(2)}$ are Wiener processes under the risk-neutral measure $\mathbb{Q}$. The log-price process $X_t = \ln S_t$ is:

$$dX_t = \left(r - \frac{v_t}{2}\right) dt + \sqrt{v_t} dW_t^{(1)}$$

Under the Euler discretisation, the discrete-time increments are:

$$\Delta X_t := X_{t+\Delta t} - X_t, \quad \Delta v_t := v_{t+\Delta t} - v_t.$$

These increments are approximately jointly normal, with mean

$$\mathbb{E}[\Delta X_t] = m_X = \left(r - \frac{1}{2}v_t\right)\Delta t, \qquad \mathbb{E}[\Delta v_t] = m_v = \kappa(\xi - v_t)\Delta t,$$

with the covariance matrix

$$\Sigma_t = \begin{pmatrix} v_t \Delta t & \rho\sigma v_t \Delta t \\ \rho\sigma v_t \Delta t & \sigma^2 v_t \Delta t \end{pmatrix} = \begin{pmatrix} \mathbb{V}\mathrm{ar}(\Delta X_t) & \mathbb{C}\mathrm{ov}(\Delta X_t, \Delta v_t) \\ \mathbb{C}\mathrm{ov}(\Delta X_t, \Delta v_t) & \mathbb{V}\mathrm{ar}(\Delta v_t) \end{pmatrix}.$$

Hence, the joint density is:

$$f(\Delta X_t, \Delta v_t, \boldsymbol{\theta}) = \frac{1}{2\pi\sqrt{\det \Sigma_t}} \exp\left(-\frac{1}{2}\left(\begin{pmatrix}\Delta X_t \\ \Delta v_t\end{pmatrix} - \begin{pmatrix} m_X \\ m_v \end{pmatrix}\right)^T \Sigma_t^{-1} \left(\begin{pmatrix}\Delta X_t \\ \Delta v_t\end{pmatrix} - \begin{pmatrix} m_X \\ m_v \end{pmatrix}\right)\right) \tag{21}$$

We can rewrite the joint density as

$$f(\Delta X_t, \Delta v_t) = f(\Delta X_t \mid \Delta v_t) f(\Delta v_t) = f(\Delta v_t \mid \Delta X_t) f(\Delta X_t).$$

For a bivariate normal vector, the corresponding conditional distributions are

$$\begin{aligned} \Delta X_t \mid \Delta v_t &\sim \mathcal{N}\left(m_X + \frac{\mathbb{C}\mathrm{ov}(\Delta X_t, \Delta v_t)}{\mathbb{V}\mathrm{ar}(\Delta v_t)}(\Delta v_t - m_v),\ \mathbb{V}\mathrm{ar}(\Delta X_t) - \frac{\mathbb{C}\mathrm{ov}(\Delta X_t, \Delta v_t)^2}{\mathbb{V}\mathrm{ar}(\Delta v_t)}\right), \\ \Delta v_t \mid \Delta X_t &\sim \mathcal{N}\left(m_v + \frac{\mathbb{C}\mathrm{ov}(\Delta X_t, \Delta v_t)}{\mathbb{V}\mathrm{ar}(\Delta X_t)}(\Delta X_t - m_X),\ \mathbb{V}\mathrm{ar}(\Delta v_t) - \frac{\mathbb{C}\mathrm{ov}(\Delta X_t, \Delta v_t)^2}{\mathbb{V}\mathrm{ar}(\Delta X_t)}\right), \end{aligned} \tag{22}$$

where

$$m_X = \left(r - \frac{1}{2}v_t\right)\Delta t, \qquad m_v = \kappa(\xi - v_t)\Delta t.$$

When calculating the priors for $\boldsymbol{\theta} = (\sigma, \kappa, \xi, \rho)$, we have to take into account the parameter constraints. Each parameter in the Heston model follows the following constraints:

$$\sigma > 0, \quad \kappa > 0, \quad \xi > 0, \quad \rho \in (-1, 1)$$

Hence, we adopt the bijections:

$$\begin{aligned}\sigma &= \exp(\theta_1),\\ \kappa &= \exp(\theta_2),\\ \xi &= \exp(\theta_3),\\ \rho &= \tanh(\theta_4),\end{aligned}$$

where each $\theta_i \in \mathbb{R}$ is unconstrained. Hence, the inverse mapping is

$$\theta_1 = \ln(\sigma), \quad \theta_2 = \ln(\kappa), \quad \theta_3 = \ln(\xi), \quad \theta_4 = \frac{1}{2}\ln\left(\frac{1+\rho}{1-\rho}\right).$$

If we want to transform the densities from $\boldsymbol{\theta}_{unc}$ back to $\boldsymbol{\theta}$, then we require the Jacobian determinant:

$$\left|\det\left(\frac{d\boldsymbol{\theta}}{d\boldsymbol{\theta}_{unc}}\right)\right| = \left|\det\begin{pmatrix} \frac{\partial\sigma}{\partial\theta_1} & 0 & 0 & 0 \\ 0 & \frac{\partial\kappa}{\partial\theta_2} & 0 & 0 \\ 0 & 0 & \frac{\partial\xi}{\partial\theta_3} & 0 \\ 0 & 0 & 0 & \frac{\partial\rho}{\partial\theta_4} \end{pmatrix}\right|.$$

This gives the following

$$\left|\det\left(\frac{d\boldsymbol{\theta}}{d\boldsymbol{\theta}_{unc}}\right)\right| = \sigma\kappa\xi(1-\rho^2).$$

Interestingly, the Jacobian matrix cancels out if we place our priors on $\boldsymbol{\theta}_{unc}$ directly, and hence does not appear in the acceptance ratio. This two-block Gibbs sampler can be broken down further by dividing the volatility path into smaller segments, so that, in each iteration, we update each individual volatility path given the parameters and data. This is known as *single-site Metropolis*.

Summary Statistics

Parameter	Mean	Standard Deviation	Effective Sample Size (ESS)
σ	0.0024	0.0252	111.6038
κ	8 297 784.4025	9 960 223.6531	2.8301
ξ	7 909 900 299 532.2783	11 693 122 264 912.1875	3.2745
ρ	-0.8547	0.3487	7.8790

Table 2: Posterior summary for the Heston parameter vector $\boldsymbol{\theta} = (\sigma, \kappa, \xi, \rho)$ under the Metropolis-within-Gibbs sampler with single-site updates of the latent volatility path. The table shows the posterior means, posterior standard deviations, and univariate ESS. The low ESS for κ, ξ, and ρ indicates substantial dependence and poor mixing in those spaces.

Figure 3 suggests that the posterior geometry is highly anisotropic across all coordinates. The σ trace remains stable on a moderate positive scale and its histogram and KDE are right-skewed, which is a direct consequence of the positive mapping $\sigma = \exp(\theta_1)$. Its ACF decays faster than for the other variables, implying comparatively better mixing for this diffusion-scale parameter. In contrast, the results suggest that κ is more tightly concentrated on [0, 0.4] and the ACF exhibits visible persistence, which, together with the relatively flat KDE over this narrow range, indicates strong local correlation and slow local moves in κ. Additionally, ξ explores values from 0 to approximately 450, with a distinctly heavy-tailed distribution and a slowly decaying ACF, indicating that the random walk behaviour is on a poorly identified scale direction. ρ is bounded between -1 and 1 but spends considerable time on the negative side. Since $\rho = \tanh(\theta_4)$, the metric compresses near ± 1, so equal-sized proposals in θ_4 translate into vanishing moves in ρ as $\rho \to 1$, producing boundary stickiness that is visible both in the trace and the ACF.

Although placing priors on the unconstrained θ cancels the Jacobian in the acceptance ratio, the mapping still induces right-skew and scale heterogeneity in (σ, κ, ξ), and strong posterior coupling between κ and ξ, which single-site state updates propagate slowly through the latent path. The result is high autocorrelation and low effective sample sizes for (κ, ξ, ρ).

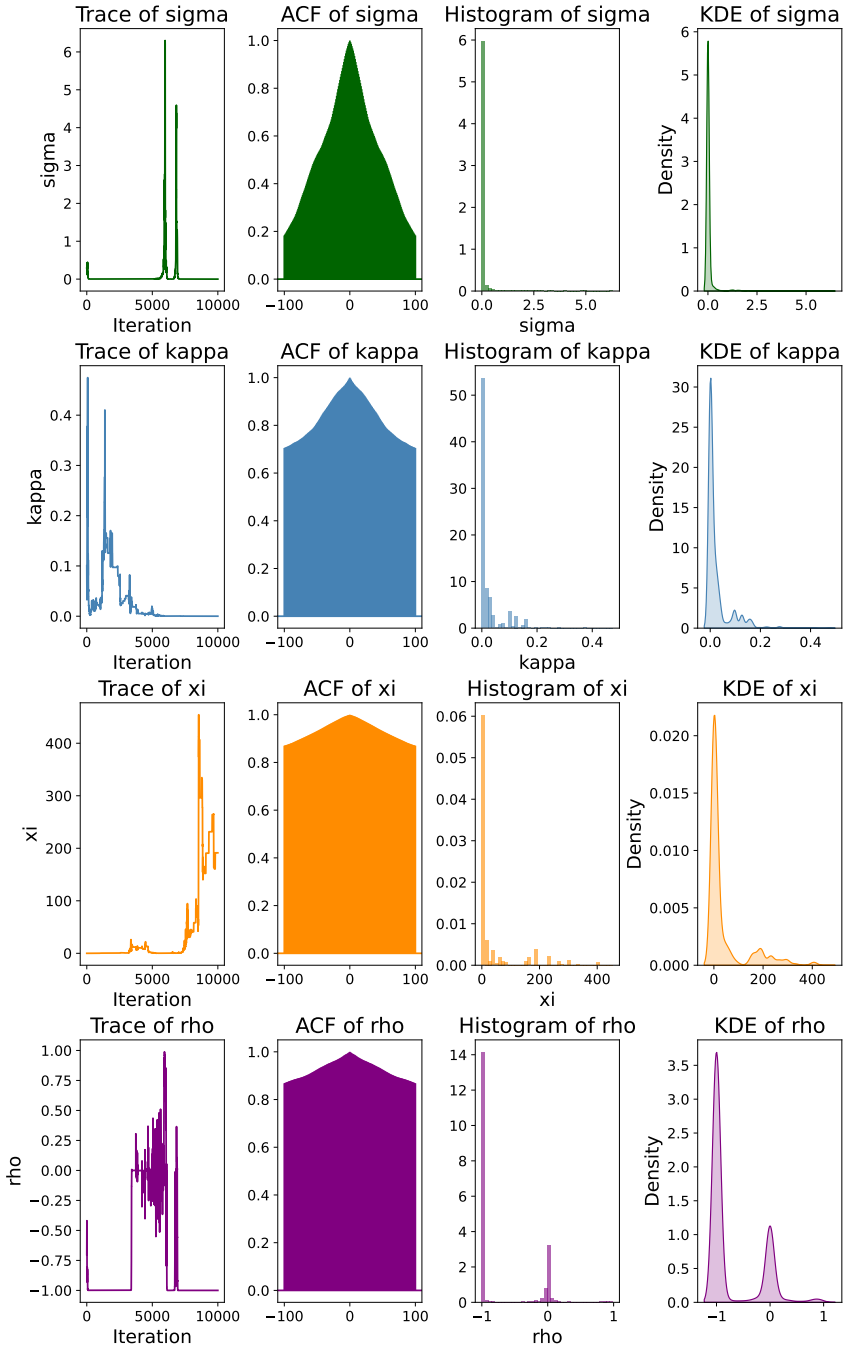


Figure 3: Diagnostic plots for each Heston parameter under the Metropolis-within-Gibbs sample: (i) trace plots, (ii) autocorrelation functions (ACF), (iii) histograms, and (iv) kernel density estimates (KDE). The σ plots show a strictly positive support with a right-skewed marginal, whereas κ is numerically compressed to the interval [0, 0.4] with a persistent autocorrelation. ξ explores a scale approximately up to 450 with a heavy right tail and ρ evolves within (-1, 1) with mass concentration on the negative side. The traces and slowly decaying ACFs indicate sluggish exploration for (κ, ξ, ρ) relative to σ.

4.3 Birth-death Markov chain Monte Carlo

In option pricing with jump-diffusions, the jump component of asset returns is naturally modelled as a count process, such as a Poisson process, to account for the occurrence of

discontinuities. A Metropolis-within-Gibbs sampler would treat the integer-valued latent counts with a proposed Gaussian random walk that moves on the count parameter or discrete jumps of arbitrary size. The former violates the integer support and requires ad hoc rounding, while the latter often leads to vanishing acceptance rates or poor mixing as the chain struggles to traverse the count space in small or overly large increments.

In the Merton jump-diffusion model, where the likelihood couples each latent jump count to both continuous diffusion residuals and a Poisson prior, these issues are even more pronounced. For example, the improper proposals distort the posterior shape, and mixing stalls in regions of high latent-count probability. To address these problems, we use a *birth-death Metropolis-Hastings* step within the Gibbs sampler. By proposing only local birth or death moves, the *birth-death Markov chain Monte Carlo (BDMCMC)* maintains detailed balance with the exact target posterior and achieves higher acceptance rates and more efficient exploration of the latent count space than ad hoc discrete updates in a standard Metropolis-within-Gibbs framework (Green, 1995).

4.3.1 Birth-death Process

A birth-death process is a continuous-time Markov chain $\{N(t)\}_{t\geq 0}$ taking values in the nonnegative integers $\mathbb{N}_0$, in which transitions may only increase the state by one (birth) or decrease by one (death). When the process is in state n, births occur at rate λ_n and deaths at rate μ_n, with $\mu_0 = 0$ by convention so that the chain cannot exit $\mathbb{N}_0$ downwards. For example, if the process represents a population, the population can never be negative. Let $P_{mn}(t) = \Pr\{N(t) = n | N(0) = m\}$, where $N(t)$ is the birth-death process. Then the forward-Kolmogorov equations are:

$$\frac{d}{dt}P_{mn}(t) = \lambda_{n-1}P_{m,n-1}(t) + \mu_{n+1}P_{m,n+1}(t) - (\lambda_n + \mu_n)P_{mn}(t),$$

with $\lambda_{-1} = 0$ and $\mu_0 = 0$. Equivalently, the infinitesimal generator $\mathcal{Q} = (q_{ij}$ has nonzero entries

$$q_{n,n+1} = \lambda_n, \quad q_{n,n-1} = \mu_n, \quad q_{nn} = -(\lambda_n + \mu_n),$$

and $q_{ij} = 0$ for $|i - j| > 1$. Since the total exit rate from state n is $\alpha_n = \lambda_n + \mu_n$, the waiting time in n is $\text{Exp}(\alpha_n)$. Hence, the chain jumps to $n+1$ with probability $\frac{\lambda_n}{\alpha_n}$ and to $n-1$ with probability $\frac{\mu_n}{\alpha_n}$. If the chain is positive recurrent, it admits a unique stationary distribution $\{\pi\}$ satisfying the detailed balance conditions:

$$\pi_n\lambda_n = \pi_{n+1}\mu_{n+1}, \quad \sum_{n\geq 0}\pi_n = 1.$$

This can be solved recursively, giving a solution for π_n:

$$\pi_n = \pi_0 \prod_{k=0}^{n-1}\frac{\lambda_k}{\mu_{k+1}}, \pi_0 = \left(1 + \sum_{n\geq 1}\prod_{k=0}^{n-1}\frac{\lambda_k}{\mu_{k+1}}\right)^{-1}.$$

4.3.2 Birth-death MCMC

As in Section 4.2, we want to sample an unobserved count $N \in \mathbb{N}_0$ from its posterior distribution $\pi(n) \propto \ell(n)p(n)$, where $\ell(n)$ is the likelihood of the data given $N = n$, and $p(n)$ is the prior probability of $N = n$. To incorporate a birth-death Metropolis-Hastings

step within a larger Gibbs sampler, we suppose that the current latent state is $N^{(i^-)} = n$ at iteration i. We then define the proposal distribution for birth and death moves:

$$b_n = \Pr\{n \to n+1\}, \quad d_n = \Pr\{n \to n-1\},$$

such that $b_n + d_n = 1$, $d_0 = 0$, and $b_n, d_n > 0$ for $n \geq 1$.[5] We then draw $U \sim \text{Uniform}(0,1)$. If $U < b_n$, we set the candidate $n^* = n+1$; otherwise $n^* = \max(n-1, 0)$. To enforce detailed balance with respect to $\pi(\cdot)$, we compute the following acceptance probability:

$$\alpha(n, n^*) = \min\left\{1, \frac{\pi(n^*)q(n^* \to n)}{\pi(n)q(n \to n^*)}\right\},$$

where $q(n \to m)$ is the proposal probability of moving from state n to state m (Robert et al., 1999). It is important to note that the birth-death Metropolis-Hastings chain is irreducible whenever $b_n > 0$ for all $n \geq 0$ and $d_n > 0$ for all $n \geq 1$, and aperiodic since there is always a positive probability of rejecting a move and remaining at n. Hence, the chain converges in distribution to the target posterior $\pi(n)$.

4.3.3 Applying Birth-death MCMC to the Merton Jump-diffusion Model

Let S_t follow the Merton risk-neutral model:

$$\frac{dS_t}{S_{t^-}} = (r - \lambda k)dt + \sigma dW_t + (e^Y - 1)dN_t,$$

where N_t is a Poisson process with rate λ and jump sizes $Y \sim \mathcal{N}(\nu, \delta^2)$ meaning that $k = \mathbb{E}[e^Y - 1]$. Over a discretisation into intervals of length Δ, let $N = N_T$ be the total count.

We aim to sample from the joint posterior of parameters and latent count N, but we focus here on updating N conditional on all else. The full conditional is $\pi(n) \propto \ell(n)p(n)$, where $\ell(N)$ is the likelihood of the observed price increments $\{S_{t_{i+1}} - S_{t_i}\}|N$, and $p(N) = \mathbb{P}(N = n) = e^{-\lambda T}\frac{(\lambda T)^n}{n!}$. At iteration i, suppose the current latent count is $N^{(i)} = n$. We draw $U \sim \text{Uniform}(0,1)$ and set the proposed count

$$n^* = \begin{cases} n+1, & U < b_n, \\ \max(n-1, 0), & U \geq b_n. \end{cases}$$

where $b_n \in (0,1)$ is the chosen birth probability and $d_n = 1 - b_n$ is the death probability. At the boundary, we set $b_0 = 1$ and $d_0 = 0$ so that negative counts are never proposed. To preserve detailed balance with respect to $\pi(n)$, we accept the move $n \to n^*$ with probability

$$\alpha = \min\left(1, \frac{\pi(n^*)q(n^* \to n)}{\pi(n)q(n \to n^*)}\right),$$

such that the proposal kernel satisfies

$$q(n \to n+1) = b_n, \quad q(n \to n-1) = d_n,$$

and likewise for $q(n^* \to n)$. Since

$$\frac{p(n+1)}{p(n)} = \frac{e^{-\lambda T}(\lambda T)^{n+1}/(n+1)!}{e^{-\lambda T}(\lambda T)^n/n!} = \frac{\lambda T}{n+1}, \quad \frac{p(n-1)}{p(n)} = \frac{n}{\lambda T},$$

[5] We set $b_n = d_n = \frac{1}{2}$ for $n \geq 1$ and $b_0 = 1$.

the prior ratio is explicit. Furthermore, under the data model, conditional on $N = n$ the log-price increment over $[0, T]$ is

$$\ln S_T - \ln S_0 = \left(r - \lambda k - \frac{1}{2}\sigma^2\right) T + \sigma W_T + \sum_{i=1}^{n} Y_i,$$

so that the continuous part and the sum of n independent, identically distributed normal variables combine into a single normal variable. Formally, for two neighbouring counts, $\ell(n) \sim \mathcal{N}(\mu_0 + n\nu, \sigma^2 T + n\delta^2)$, where $\mu_0 := (r - \lambda k - \frac{1}{2}\sigma^2)T$. Thus

$$\frac{\ell(n+1)}{\ell(n)} = \frac{\varphi(x; \mu_0 + (n+1)\nu, \sigma^2 T + (n+1)\delta^2)}{\varphi(x; \mu_0 + n\nu, \sigma^2 T + n\delta^2)},$$

where $\varphi(x; m, s^2)$ denotes the normal density with mean m and variance s^2:

$$\varphi(x; m, s^2) = \frac{1}{\sqrt{2\pi s^2}} \exp\left(-\frac{(x-m)^2}{2s^2}\right).$$

Hence the birth move $n \to n+1$ is

$$\alpha_b = \min\left(1, \frac{\ell(n+1)}{\ell(n)} \frac{\lambda T}{n+1} \frac{d_{n+1}}{b_n}\right),$$

and the death move $n \to n-1$ is

$$\alpha_d = \min\left(1, \frac{\ell(n-1)}{\ell(n)} \frac{n}{\lambda T} \frac{b_{n-1}}{d_n}\right).$$

Summary Statistics

Parameter	Mean	Standard Deviation	Effective Sample Size (ESS)
σ	-0.0092	0.0204	32.473714
λ	0.2381	0.2946	4.530036
ν	0.4606	0.1980	3.371626
δ	-0.5295	0.2700	2.495140

Table 3: Posterior summary for the Birth-death MCMC sampler in the jump-diffusion setting, for $\boldsymbol{\theta} = (\sigma, \lambda, \nu, \delta)$, showing posterior mean, posterior standard deviation, and univariate ESS. The low ESS for λ, ν, δ relative to σ shows dependence in the jump component, consistent with strong posterior coupling between the jump and jump-size law.

Figure 4 shows different behaviours across coordinates. For σ, the chain shows frequent level crossings and relatively short-lived local persistence, indicating that the diffusion scale is identified through the continuous component of returns. In contrast, the jump parameters display infrequent regime switches, with the path remaining near a local level for extended stretches before moving to a new region. This stickiness is a structural consequence of the discrete latent count n and the acceptance probabilities for the birth/death moves. When n is near its posterior mode, the likelihood ratio disfavors moves that push the chain away from that mode. When n is far away, the proposal often attempts

low-probability corrections. In both cases, local proposals in n pass inertia to (λ, ν, δ) through the tight coupling between the jump frequency and jump sizes. The effect is a trace with plateau-like segments and delayed mean reversion for the jump parameters, which is consistent with the low ESS reported in the table.

Additionally, the ACF reinforces this picture. σ has a faster decay, whereas λ, ν, δ retain substantial correlation across lags. The marginal for σ is comparatively concentrated and near-Gaussian on the working scale. In contrast, λ is right-skewed with mass near low intensities and a diffuse tail, while ν and δ are broader and skewed, reflecting compensation between jump size location and scale. In other words, increases in ν can be partially offset by decreases in δ, and vice versa, to produce comparable aggregate jump contributions. Hence, the ridge-like coupling naturally widens the KDes and thickens tails.

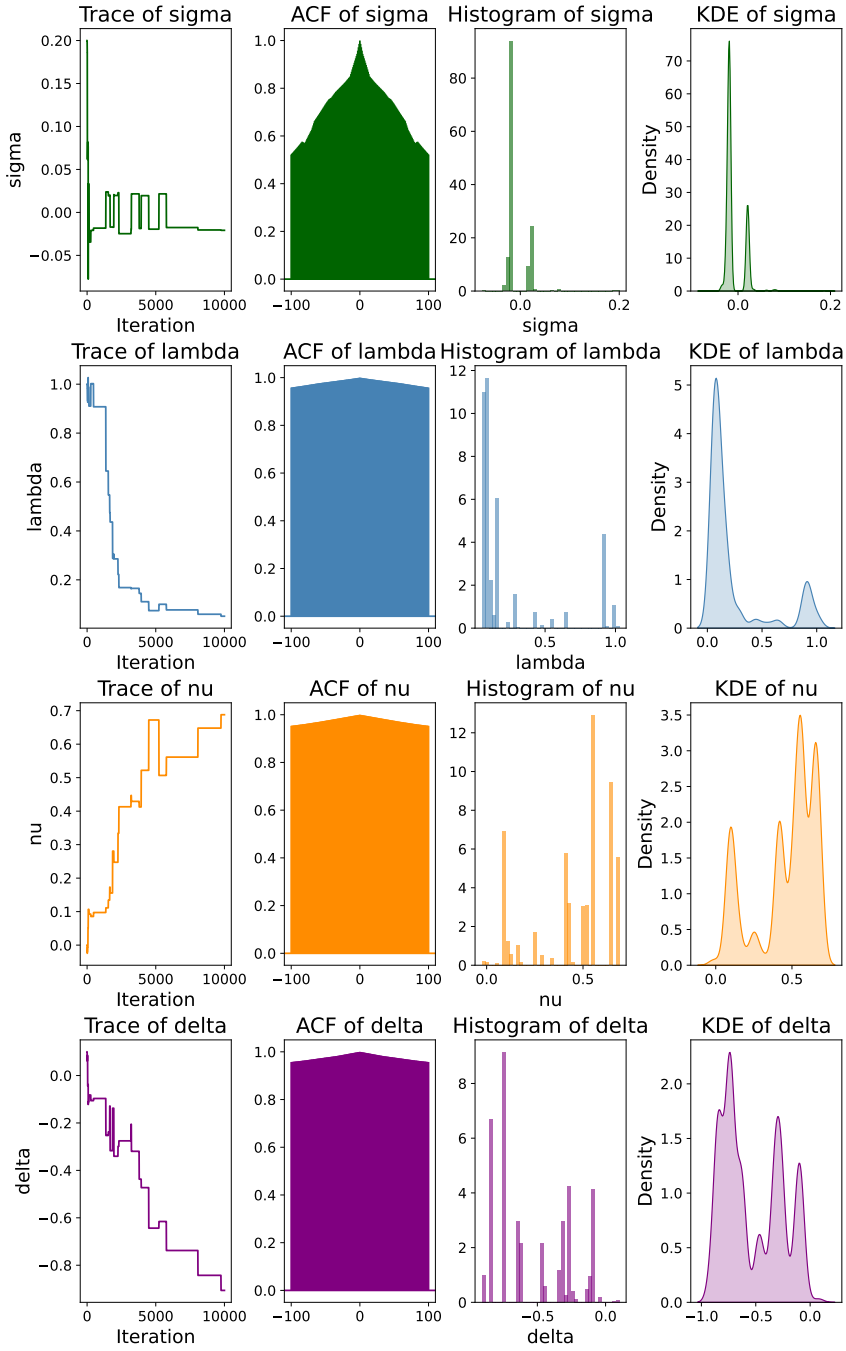


Figure 4: Diagnostic plots for the Birth-death MCMC output by parameter $\boldsymbol{\theta} = (\sigma, \lambda, \nu, \delta)$. The traces for λ, ν, δ display long visits and slow regime changes. Their ACFs decay slowly, and the distributions are skewed with broad dispersion.

Practically, birth-death MCMC can identify large jumps, but jump-count recovery requires sufficiently informative data. In applications, this means that higher-frequency returns, stronger priors, or additional option-implied information may be needed to infer jump intensity reliably. For this paper, the birth-death experiment justifies the method and has demonstrated the difficulty of latent jump inference.

4.4 Hamiltonian Monte Carlo

Up to this point, we have struggled to get good results for complex pricing models such as Heston and Merton jump-diffusion models. In many Bayesian inference problems, standard random-walk Metropolis or Gibbs samplers suffer from poor scaling in high-dimensional parameter spaces or when strong posterior correlations exist. *Hamiltonian Monte Carlo (HMC)* addresses these limitations by introducing *auxiliary momentum* variables and simulating Hamiltonian dynamics to propose moves that preserve detailed balance. The main idea is to view sampling from a target density $\pi(\boldsymbol{\theta})$ as exploring the level sets of an energy function, and to exploit gradient information to traverse parameter space efficiently (Duane et al., 1987).

Formally, let $\boldsymbol{\theta} \in \mathbb{R}^D$ denote the vector of parameters whose posterior density is $\pi(\boldsymbol{\theta}) \propto e^{-U(\boldsymbol{\theta})}$, where

$$U(\boldsymbol{\theta}) = -\log[\pi(\boldsymbol{\theta})]$$

is called the potential energy. We then introduce an auxiliary momentum variable $p \in \mathbb{R}^D$with a conditional Gaussian density $p \sim \mathcal{N}(0, M)$, where M is a positive-define *mass matrix*. The joint density over (θ, p) is then

$$\Pi(\boldsymbol{\theta}, p) = \pi(\boldsymbol{\theta})\mathcal{N}(p|0, M) \propto \exp(-H(\boldsymbol{\theta}, p)),$$

with Hamiltonian

$$H(\boldsymbol{\theta}, p) = U(\boldsymbol{\theta}) + K(p), \quad K(p) = \frac{1}{2}p^\top M^{-1}p,$$

the kinetic energy.

We can describe how $(\boldsymbol{\theta}, p)$ evolve in continuous time t through the *Hamilton's equations of motion*:

$$\frac{d\boldsymbol{\theta}}{dt} = \frac{\partial H}{\partial p} = M^{-1}p, \quad \frac{dp}{dt} = -\frac{\partial H}{\partial \boldsymbol{\theta}} = -\nabla U(\boldsymbol{\theta}). \tag{23}$$

These deterministic dynamics preserve the joint density $\Pi(\boldsymbol{\theta}, p)$, i.e, they conserve H, and are volume-preserving in phase space. Since the exact integration of (23) is generally impossible, we apply a symplectic integrator, usually *leapfrog scheme*, to approximate the flow with small step size ϵ and repeating the following three steps L times to yield a proposal $(\boldsymbol{\theta}^*, p^*)$:

$$\begin{aligned}
p\left(t + \frac{\epsilon}{2}\right) &= p(t) - \frac{\epsilon}{2}\nabla U(\boldsymbol{\theta}(t)) \qquad \text{(Half-momentum update)}, \\
\boldsymbol{\theta}(t + \epsilon) &= \boldsymbol{\theta}(t) + \epsilon M^{-1} p\left(t + \frac{\epsilon}{2}\right) \qquad \text{(Position update)}, \\
p(t + \epsilon) &= p\left(t + \frac{\epsilon}{2}\right) - \frac{\epsilon}{2}\nabla U(\boldsymbol{\theta}(t + \epsilon)) \qquad \text{Half-momentum update)}.
\end{aligned}$$

As numerical integration incurs discretisation error, the Hamiltonian is no longer exactly conversed, and to correct for this and to ensure detailed balance, we perform a Metropolis acceptance step to accept $(\boldsymbol{\theta}^*, p^*)$ with probability

$$\min\{1, \exp[-H(\boldsymbol{\theta}^*, p^*) + H(\boldsymbol{\theta}, p)]\}$$

Finally, new momentum is resampled from $\mathcal{N}(0, M)$ before commencing the next trajectory. HMC can propose moves that travel in parameter space with high probability by using gradient information, reducing the random-walk behaviour that simpler MCMC methods use.

4.4.1 Applying HMC to the Heston Model

Recall that the Heston model under the risk-neutral measure is

$$\begin{aligned} dS_t &= rS_t dt + S_t\sqrt{v_t}dW^{\mathbb{Q}}_{t,1}, \\ dv_t &= \kappa(\xi - v_t)dt + \sigma\sqrt{v_t}dW^{\mathbb{Q}}_{t,2}, \\ d\langle W^{\mathbb{Q}}_{t,1}, W^{\mathbb{Q}}_{t,2}\rangle &= \rho dt. \end{aligned}$$

For simplicity, we will write $W^{\mathbb{Q}}_{t,i} := W_{t,i}$ for $i = 1, 2$. To apply HMC to Heston, we require defining a suitable potential energy function and computing its gradient with respect to the model parameters $\boldsymbol{\theta} = (\kappa, \xi, \sigma, \rho)$ and the latent volatility path $\{v_t\}$. Given discrete observations $S_{t_0}, S_{t_1}, \ldots, S_{t_N}$ and latent variances $v_0, v_1, \ldots, v_N$, define

$$\Delta Y_i = \begin{pmatrix} \Delta \log S_i \\ \Delta v_i \end{pmatrix}, \qquad \Delta \log S_i = \log\left(\frac{S_i}{S_{i-1}}\right), \qquad \Delta v_i = v_i - v_{i-1}.$$

Under the Euler discretisation, the Heston increments are approximately jointly normal:

$$\Delta Y_i \mid v_{i-1}, \boldsymbol{\theta} \sim \mathcal{N}\left(\boldsymbol{m}_i(\boldsymbol{\theta}), \Sigma_i(\boldsymbol{\theta})\right),$$

where

$$\boldsymbol{m}_i(\boldsymbol{\theta}) = \begin{pmatrix} \left(r - \frac{1}{2}v_{i-1}\right)\Delta t \\ \kappa(\xi - v_{i-1})\Delta t \end{pmatrix},$$

and

$$\Sigma_i(\boldsymbol{\theta}) = v_{i-1}\Delta t \begin{pmatrix} 1 & \rho\sigma \\ \rho\sigma & \sigma^2 \end{pmatrix}.$$

Therefore, the log-likelihood contribution of the i-th increment is

$$\mathcal{L}_i(\boldsymbol{\theta}) = -\frac{1}{2}\left[\log\left((2\pi)^2 \det \Sigma_i(\boldsymbol{\theta})\right) + \left(\Delta Y_i - \boldsymbol{m}_i(\boldsymbol{\theta})\right)^\top \Sigma_i(\boldsymbol{\theta})^{-1}\left(\Delta Y_i - \boldsymbol{m}_i(\boldsymbol{\theta})\right)\right].$$

This Euler likelihood is only well-defined for positive latent variances. Hence the joint posterior can be written as

$$\pi(\boldsymbol{\theta}, \{v_i\} \mid \{S_i\}) \propto \pi(\boldsymbol{\theta})\pi(v_0)\prod_{i=1}^{N}\exp\left(\mathcal{L}_i(\boldsymbol{\theta})\right),$$

and the corresponding potential energy is

$$U(\boldsymbol{\theta}, \{v_i\}) = -\log\pi(\boldsymbol{\theta}) - \log\pi(v_0) - \sum_{i=1}^{N}\mathcal{L}_i(\boldsymbol{\theta}).$$

Gradient components are then obtained by differentiating this potential energy with respect to each parameter and each latent variance v_i. In practice, these gradients may be computed analytically or by automatic differentiation, and are then assembled into $\nabla_{\boldsymbol{\theta},v}U$ for the leapfrog updates.

Figure 5 shows reduced random-walk behaviour, stationary traces, and largely unimodal marginals, which is expected with HMC. The mean reversion (κ) and long-run variance (ξ) show the cleanest diagnostics, with short-memory traces and sharply decaying ACFs with tight KDEs. This is reflected in the ESS results in Table 4. In contrast, the volatility

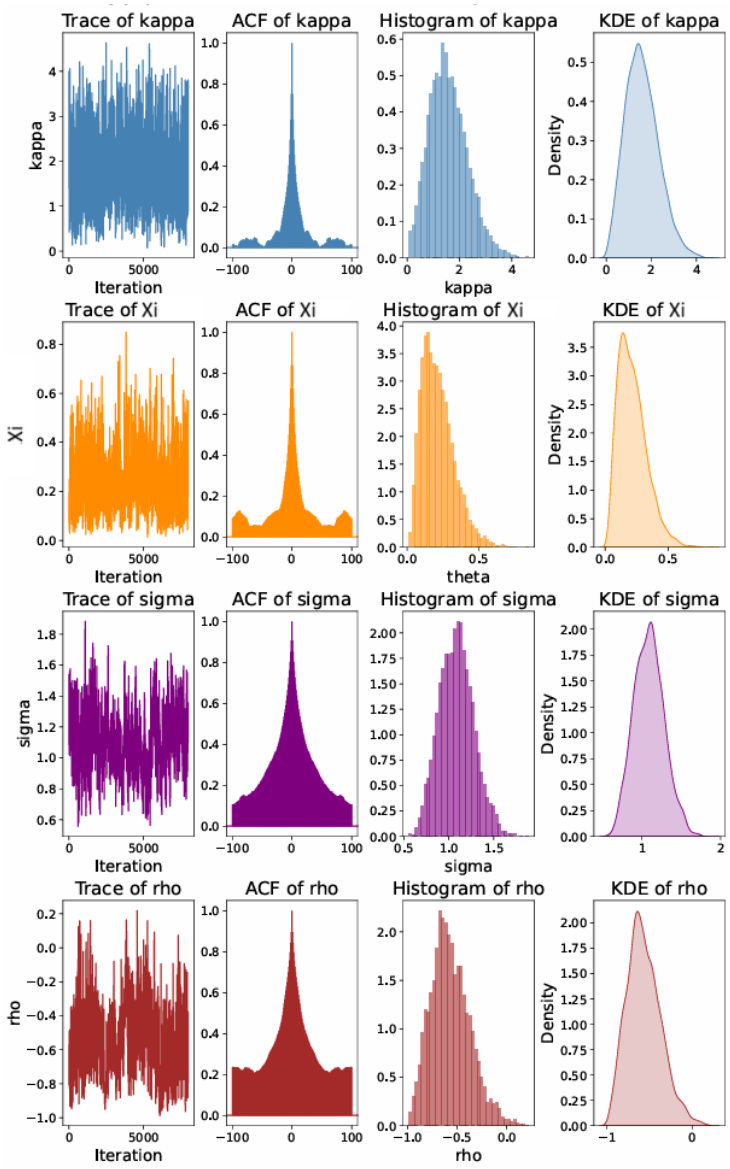


Figure 5: HMC diagnostic plots for the Heston model by parameter $\boldsymbol{\theta} = (\kappa, \xi, \sigma, \rho)$: (i) trace plots, (ii) autocorrelation functions, (iii) histograms, and (iv) kernel density estimates. The traces are stationary with frequent level crossings and the ACFs for κ and ξ decay rapidly, whereas σ and ρ exhibit slower ACF decay. The marginal distributions are unimodal, with broader uncertainty for σ and negative concentration for ρ.

HMC Heston: Summary Statistics

Parameter	Mean	Standard Deviation	Effective Sample Size (ESS)
κ	1.5888	0.7302	362.592367
ξ	0.2174	0.1135	220.770254
σ	1.0890	0.1922	91.640614
ρ	-0.5650	0.1991	50.603245

Table 4: Posterior summary for the Heston parameters under Hamiltonian Monte Carlo. The table shows posterior means, posterior standard deviations, and univariate ESS for $\boldsymbol{\theta} = (\kappa, \xi, \sigma, \rho)$. The ESS values show efficient exploration for κ and ξ, with comparatively slower mixing for σ and ρ.

of volatility (σ) and leverage (ρ) retain longer memory. σ has a broader marginal and a noticeable ACF tail, while ρ concentrates on negative values with slower ACF decay.

Gradient-informed proposals allow long, high-acceptance trajectories along the principal directions of (κ, ξ), mitigating the ridge structure that hampers random-walk samplers. Hence, these parameters have rapid ACF decay and larger ESS. The σ parameter couples to both the latent variance dynamics and the scaling of the price model, producing a stiffer curvature and wider marginal. The finite step size and generic mass matrix leave some residual autocorrelation, thereby lowering ESS. For $\rho \in (-1, 1)$, the parameterisation induces geometric compression as $|\rho| \to 1$, so equal momentum updates translate into smaller moves near the boundaries. In other words, on the unconstrained Fisher-z scale, $\rho = \tanh(\theta_4)$, and

$$\frac{d\rho}{d\theta_4} = 1 - \rho^2,$$

where $\boldsymbol{\theta}_{unc} := (\theta_1, \theta_2, \theta_3, \theta_4)$ denotes the unconstrained parameters of the model. Thus, a fixed-size update on θ_4 induces a change in ρ of order $(1 - \rho^2)\Delta\theta_4$. As $|\rho| \to 1$, this factor vanishes, geometrically compressing moves near the boundaries.

4.4.2 Applying HMC to the Multi-Asset Black-Scholes Model

Recall that the multi-asset Black-Scholes model under the risk-neutral measure is

$$\begin{aligned} dS_{1,t} &= rS_{1,t}dt + \sigma S_{1,t}dW_{1,t}, \\ dS_{2,t} &= rS_{2,t}dt + \nu S_{2,t}dW_{2,t}, \\ d\langle W_1, W_2\rangle_t &= \rho dt. \end{aligned}$$

To apply HMC to the multi-asset Black-Scholes model, we need to define a suitable potential energy function and compute its gradient with respect to the model parameters $\boldsymbol{\theta} = (\sigma, \nu, \rho)$. Suppose observations are available at times $0 = t_0 < t_1 < \cdots < t_N$ with uniform step $\Delta t = t_i - t_{i-1}$. Define the bivariate log-return increment $\Delta X_i := (\Delta X_{1,i}, \Delta X_{2,i})^\top$ with $\Delta X_{1,i} = X_{1,i} - X_{1,i-1} = \log\left(\frac{S_{1,t_i}}{S_{1,t_{i-1}}}\right)$ and $\Delta X_{2,i} = X_{2,i} - X_{2,i-1} = \log\left(\frac{S_{2,t_i}}{S_{2,t_{i-1}}}\right)$. As in Section 4.2, we can use the Euler-Maruyama approximation to get Gaussian increments with mean

$$\boldsymbol{\mu}(\boldsymbol{\theta}) = \left(\left(r - \frac{1}{2}\sigma^2\right)\Delta t, \left(r - \frac{1}{2}\nu^2\right)\Delta t\right)^\top,$$

and covariance

$$\Sigma(\boldsymbol{\theta}) = \Delta t \begin{pmatrix} \sigma^2 & \rho\sigma\nu \\ \rho\sigma\nu & \nu^2 \end{pmatrix},$$

where $\boldsymbol{\theta} := (\sigma, \nu, \rho)$. Writing the residuals as $\boldsymbol{z}_i(\boldsymbol{\theta}) := \Delta \boldsymbol{X}_i - \boldsymbol{\mu}(\boldsymbol{\theta})$, each increment likelihood can be written as the bivariate normal density

$$\ell(\Delta \boldsymbol{X}_i|\boldsymbol{\theta}) = \frac{1}{2\pi\sqrt{\det \Sigma(\boldsymbol{\theta})}} \exp\left(-\frac{1}{2}\boldsymbol{z}_i(\boldsymbol{\theta})^\top \Sigma(\boldsymbol{\theta})^{-1} \boldsymbol{z}_i(\boldsymbol{\theta})\right).$$

Hence the log likelihood for the entire series is

$$\mathcal{L}(\boldsymbol{\theta}) = \sum_{i=1}^{N} \log \ell(\Delta \boldsymbol{X}_i|\boldsymbol{\theta}) = -\frac{N}{2}\log((2\pi)^2) - \frac{N}{2}\log\det\Sigma(\boldsymbol{\theta}) - \frac{1}{2}\sum_{i=1}^{N} \boldsymbol{z}_i(\boldsymbol{\theta})^\top \Sigma(\boldsymbol{\theta})^{-1}\boldsymbol{z}_i(\boldsymbol{\theta}).$$

Let $\pi(\boldsymbol{\theta})$ denote a prior density. The potential energy for HMC is the negative log-posterior:

$$U(\boldsymbol{\theta}) = -\log \pi(\boldsymbol{\theta}) - \mathcal{L}(\boldsymbol{\theta}),$$

and the HMC force is $\nabla_{\boldsymbol{\theta}} U(\boldsymbol{\theta}) = -\nabla_{\boldsymbol{\theta}} \log \pi(\boldsymbol{\theta}) - \nabla_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta})$. Recall that:

$$\det \Sigma(\boldsymbol{\theta}) = \Delta t^2 \sigma^2 \nu^2 (1-\rho^2), \quad \Sigma(\boldsymbol{\theta})^{-1} = \frac{1}{\Delta t(1-\rho^2)} \begin{pmatrix} \sigma^{-2} & -\rho(\sigma\nu)^{-1} \\ -\rho(\sigma\nu)^{-1} & \nu^{-2}. \end{pmatrix}$$

To compute $\nabla_{\boldsymbol{\theta}} \mathcal{L}$, we begin by differentiating the following terms:

$$\begin{aligned} \frac{\partial}{\partial \vartheta}(\boldsymbol{z}^\top \Sigma^{-1} \boldsymbol{z}) &= -2(\partial_\vartheta \boldsymbol{\mu})^\top \Sigma^{-1}\boldsymbol{z} - \boldsymbol{z}^\top \Sigma^{-1}(\partial_\vartheta \Sigma)\Sigma^{-1}\boldsymbol{z}, \\ \frac{\partial}{\partial \vartheta}\log\det\Sigma &= \operatorname{tr}(\Sigma^{-1}\partial_\vartheta \Sigma), \end{aligned} \tag{24}$$

for $\vartheta \in \{\sigma, \nu, \rho\}$. The derivatives of the mean and covariance for each parameter are

$$\partial_\sigma \boldsymbol{\mu} = (-\sigma\Delta t, 0)^\top, \quad \partial_\nu \boldsymbol{\mu} = (0, -\nu\Delta t)^\top, \quad \partial_\rho \boldsymbol{\mu} = (0,0)^\top, \tag{25}$$

and

$$\partial_\sigma \Sigma = \Delta t \begin{pmatrix} 2\sigma & \rho\nu \\ \rho\nu & 0 \end{pmatrix}, \quad \partial_\nu \Sigma = \Delta t \begin{pmatrix} 0 & \rho\sigma \\ \rho\sigma & 2\nu \end{pmatrix}, \quad \partial_\rho \Sigma = \Delta t \begin{pmatrix} 0 & \sigma\nu \\ \sigma\nu & 0 \end{pmatrix}. \tag{26}$$

From $\det\Sigma$, we can see that

$$\partial_\sigma \log\det\Sigma = \frac{2}{\sigma}, \quad \partial_\nu \log\det\Sigma = \frac{2}{\nu}, \quad \partial_\rho \log\det\Sigma = -\frac{2\rho}{1-\rho^2}. \tag{27}$$

Substituting (24), (25), (26), and (27) into $\partial_{\boldsymbol{\theta}}\mathcal{L}(\boldsymbol{\theta})$ we get, for each $\vartheta \in \{\sigma, \nu, \rho\}$,

$$\partial_\vartheta \mathcal{L}(\boldsymbol{\theta}) = -\frac{N}{2}\partial_\vartheta \log\det\Sigma(\boldsymbol{\theta}) + \sum_{i=1}^{N}(\partial_\vartheta \boldsymbol{\mu})^\top \Sigma(\boldsymbol{\theta})^{-1}\boldsymbol{z}_i(\boldsymbol{\theta}) + \frac{1}{2}\sum_{i=1}^{N} \boldsymbol{z}_i(\boldsymbol{\theta})^\top \Sigma(\boldsymbol{\theta})^{-1}(\partial_\vartheta \Sigma(\boldsymbol{\theta}))\Sigma(\boldsymbol{\theta})^{-1}\boldsymbol{z}_i(\boldsymbol{\theta}).$$

For numerical stability and constraint handling, we work on an unconstrained parameter vector $\boldsymbol{\theta}_{unc} = (\theta_1, \theta_2, \theta_3) \in \mathbb{R}^3$, as before, via the smooth bijection $\sigma = \exp(\theta_1), \nu = \exp(\theta_2)$, and $\rho = \tanh(\theta_3)$. If priors are placed directly on $\boldsymbol{\theta}_{unc}$, then the log-Jacobian does not enter the acceptance ratio, and the chain rule gives

$$\frac{\partial U}{\partial \theta_1} = \frac{\partial U}{\partial \sigma}\sigma, \quad \frac{\partial U}{\partial \theta_2} = \frac{\partial U}{\partial \nu}\nu, \quad \frac{\partial U}{\partial \theta_3} = \frac{\partial U}{\partial \rho}(1-\rho^2).$$

However, if the priors are specified on the constrained parameters (σ, ν, ρ), then the posterior density on the unconstrained space must include the Jacobian $J(\boldsymbol{\theta}_{unc}) = \sigma\nu(1-\rho^2)$. Equivalently, if $U_c(\sigma, \nu, \rho) = -\log \pi(\sigma, \nu, \rho) - \mathcal{L}(\sigma, \nu, \rho)$ denotes the constrained-space potential energy, then the unconstrained-space potential energy is

$$U_{unc}(\theta_1, \theta_2, \theta_3) = U_c(\sigma, \nu, \rho) - \log\left[\sigma\nu(1-\rho^2)\right],$$

where $\sigma = e^{\theta_1}$, $\nu = e^{\theta_2}$, $\rho = \tanh(\theta_3)$. Therefore, the chain-rule derivatives become

$$\frac{\partial U_{unc}}{\partial \theta_1} = \sigma \frac{\partial U_c}{\partial \sigma} - 1,$$
$$\frac{\partial U_{unc}}{\partial \theta_2} = \nu \frac{\partial U_c}{\partial \nu} - 1,$$

and

$$\frac{\partial U_{unc}}{\partial \theta_3} = (1-\rho^2)\frac{\partial U_c}{\partial \rho} + 2\rho.$$

We then introduce a momentum variable $p \sim \mathcal{N}(0, M)$ with positive-definite mass matrix M, which defines the Hamiltonian $H(\boldsymbol{\theta}_{unc}, p) = U(\boldsymbol{\theta}_{unc}) + \frac{1}{2}p^\top M^{-1} p$. Hamilton's equations are then

$$\frac{d\boldsymbol{\theta}_{unc}}{dt} = M^{-1}p, \quad \frac{dp}{dt} = -\nabla_{\boldsymbol{\theta}_{unc}} U(\boldsymbol{\theta}_{unc}).$$

The leapfrog integrator with step size $\epsilon > 0$ and L steps implements a half step momentum update using $-\nabla_{\boldsymbol{\theta}_{unc}} U$, a full-step position update via $M^{-1}p$, and a final half-step momentum update. The proposal $(\boldsymbol{\theta}^*_{unc}, p^*)$ is accepted with probability

$$\alpha = \min\left\{1, \exp(-H(\boldsymbol{\theta}^*_{unc}, p^*) + H(\boldsymbol{\theta}_{unc}, p))\right\}.$$

HMC Multi Black-Scholes: Summary Statistics

Parameter	Mean	Standard Deviation	Effective Sample Size (ESS)
σ	0.3205	0.0144	5161.588590
ν	0.2460	0.0110	5378.332800
ρ	0.5189	0.0459	5195.554648

Table 5: Posterior summary for the Multi Black-Scholes parameters under Hamiltonian Monte Carlo. The table shows posterior means, posterior standard deviations, and univariate ESS for $\boldsymbol{\theta} = (\sigma, \nu, \rho)$. The ESS values indicate highly efficient exploration and very weak correlation in the draws, with tight marginal uncertainty for σ and ν and moderate dispersion for ρ.

The diagnostic plots in Figure 6 show an efficiently tuned Hamiltonian Monte Carlo sampler. Trajectories traverse level sets of the posterior with minimal random-walk behaviour, creating stationary traces with dense coverage of the support and ACFs that fall to zero after a few lags. The distributions of σ and ν are near-Gaussian, consistent with the small posterior standard deviations in Table 5, and ρ exhibits a wider but still unimodal marginal with the trace remaining comfortably away from the boundaries, which is reflected in a slightly broader distribution yet comparably fast ACF decay.

When we compare these results to the results in Section 4.1.1, HMC attains significantly higher ESS and visibly faster ACF decay for each coordinate. In Figure 2, the ACFs decay more slowly and the traces show longer plateaus, particularly for the correlation coordinate, whereas HMC's momentum-based proposals generate moves aligned with the posterior curvature, suppressing diffusive behaviour and reducing integrated autocorrelation time. In addition, the leapfrog integrator preserves a constant Hamiltonian along trajectories, delivering high acceptance without shrinking step sizes, whereas the adaptive random walk in Figure 2 must balance acceptance against step length, which leads to shorter effective moves in high-curvature directions. Hence, for the same nominal iteration budget, HMC produces larger ESS with narrower distributions for all parameters.

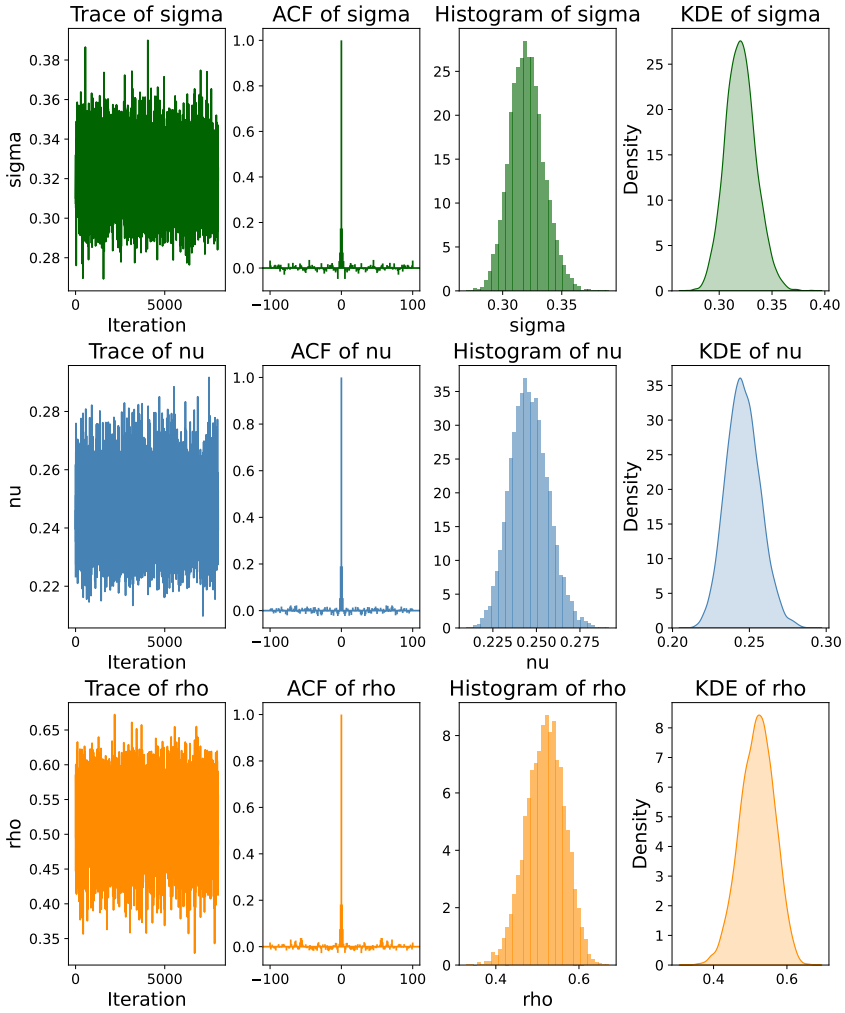


Figure 6: HMC diagnostic plots for the multi-asset Black-Scholes model by parameter $\boldsymbol{\theta} = (\sigma, \nu, \rho)$: (i) trace plots, (ii) autocorrelation functions, (iii) histograms, and (iv) kernel density estimates. The traces show stationary exploration with frequent level crossings, and the ACFs decay to near zero at very low lags. Distributions are unimodal and concentrated for σ and ν, and moderately broader for ρ.

5 Experiments and Results

The experiments in this section are deliberately designed as controlled synthetic studies. Reliable historical data for exotic options, especially path-dependent contracts such as Asian options, is not generally available in the same way as listed vanilla option data. Many exotic derivatives are traded over the counter, and their contract specifications, monitoring conventions, and historical transaction prices are not publicly observable with sufficient consistency for transparent calibration. Hence, we use a synthetic design to avoid this ambiguity and ensure that the data-generating model and true parameter values are known.

The experiments are organised into three groups. The first group consists of validation experiments to check whether the samples behave as expected. The second group consists of correctly specified exotic option calibration experiments, in which the model used to generate synthetic option prices is also the model used for calibration. These experiments test whether MCMC can recover parameters and reproduce the synthetic option price surface. The final group consists of a misspecification experiment, in which Heston-generated Asian option prices are fitted using a Merton jump-diffusion model.

Throughout this section, a synthetic observed option price is generated as

$$y_i = C_i(\boldsymbol{\theta}^\star) + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0, \tau^2), \qquad i = 1, \ldots, n,$$

where $C_i(\boldsymbol{\theta}^\star)$ is the model-implied price of the i-th contract under the true data-generating parameter vector $\boldsymbol{\theta}^\star$. The parameter τ is the synthetic observation-noise level. The contracts are arranged across strikes and maturities, so that each calibration problem is a surface-fitting problem rather than a single-price fit.

For each experiment, posterior performance is assessed using three types of diagnostics. First, posterior marginal densities are compared with the true parameter values when such values exist. Second, posterior predictive means are compared with the synthetic observed prices. Third, residual heatmaps are used to detect systematic structure across strike and maturity. The residual for contract i is defined by $e_i = y_i - \widehat{C}_i$, where

$$\widehat{C}_i = \mathbb{E}\left[C_i(\boldsymbol{\theta}) \mid \boldsymbol{y}\right]$$

is approximated by the posterior predictive mean. Positive residuals therefore indicate that the fitted model underprices the synthetic observation, while negative residuals indicate that the fitted model overprices it.

The experiments used in this section are summarised in Table **??**. The first experiment calibrates a one-dimensional Black-Scholes geometric Asian model using random-walk Metropolis. The second calibrates a two-asset Black-Scholes Margrabe model using adaptive Metropolis-Hastings. The third tests birth-death MCMC on a synthetic Merton jump-diffusion return path. The fourth and fifth experiments calibrate Merton and Heston geometric Asian option models under correct specification. The final experiment fits a Merton jump-diffusion model to Heston-generated Asian option prices.

5.1 Black-Scholes Geometric Asian Option

The first experiment validates the basic random-walk Metropolis implementation in a one-dimensional setting. Synthetic geometric Asian option prices are generated under

Experiment	Data-generating model	Fitted model	Main sampler
Black-Scholes Asian	Black-Scholes	Black-Scholes	Random-walk MH
Margrabe exchange option	Two-asset Black-Scholes	Two-asset Black-Scholes	Adaptive MH
Merton Asian	Merton jump-diffusion	Merton jump-diffusion	MH within Gibbs
Heston Asian	Heston	Heston	HMC
Misspecification	Heston	Merton jump-diffusion	MH within Gibbs

Table 6: Summary of experiments.

the Black-Scholes model using the analytical formula derived earlier in the paper. The risk-neutral dynamics are $\frac{dS_t}{S_t} = r\,dt + \sigma^\star\,dW_t$, where $\sigma^\star = 0.22$ is the true volatility. The calibration model is correctly specified and estimates only σ. Positivity is enforced by the transformation $\sigma = \exp(\eta_\sigma)$.

The posterior summary is given in Table **??**. The posterior mean is 0.2238, close to the true value 0.22, and the 95% credible interval is approximately $[0.2165, 0.2312]$. The true value lies well inside this interval. The effective sample size is high relative to the length and computational cost of the run, which is expected because the posterior is one-dimensional and the pricing formula is analytical.

Parameter	True	Mean	SD	2.5%	Median	97.5%
σ	0.2200	0.2238	0.0039	0.2165	0.2238	0.2312

Table 7: Posterior summary for the Black-Scholes geometric Asian experiment.

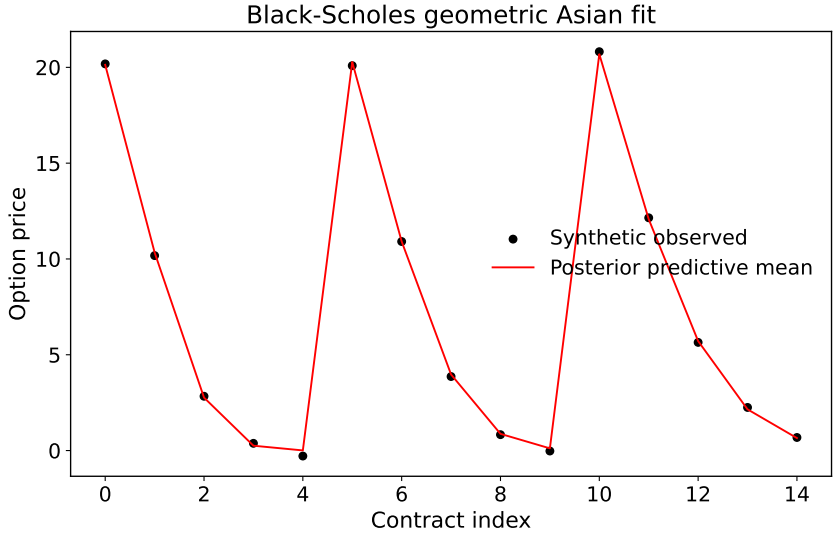


Figure 7: Posterior predictive fit for the Black-Scholes geometric Asian option experiment. Black points denote synthetic observed prices and the red line denotes the posterior predictive mean.

Figure 7 compares the synthetic observed prices with the posterior predictive mean. The two series are nearly indistinguishable at the plotted scale. The contract index orders

options across strike and maturity combinations, so the sharp changes in the line reflect movements across the option panel rather than a time-series pattern.

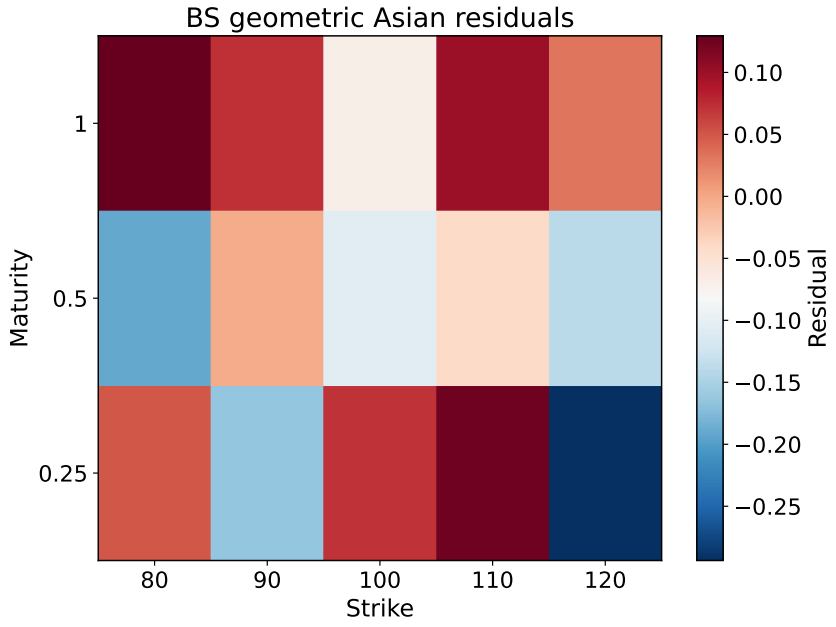


Figure 8: Residual heatmap for the Black-Scholes geometric Asian option experiment.

Figure 8 shows residuals across strike and maturity. Mathematically, the residual heatmap is a diagnostic for $\mathbb{E}\left[y_i - C_i(\sigma) \mid \boldsymbol{y}\right]$. The residuals are small relative to the scale of the option prices, and there is no dominant monotone pattern across the surface. Hence, this supports the conclusion that the model is correctly specified and that the posterior predictive mean explains the synthetic prices without systematic distortion.

5.2 Black-Scholes Margrabe Exchange Option

The second experiment examines adaptive Metropolis-Hastings in a multi-asset setting. Synthetic exchange option prices are generated under the two-asset Black-Scholes model,

$$\frac{dS_{1,t}}{S_{1,t}} = r\,dt + \sigma_1^\star\,dW_{1,t}, \qquad \frac{dS_{2,t}}{S_{2,t}} = r\,dt + \sigma_2^\star\,dW_{2,t},$$

where $d\langle W_1, W_2\rangle_t = \rho^\star\,dt$. The true values are $\sigma_1^\star = 0.20, \sigma_2^\star = 0.28,$ and $\rho^\star = 0.45$.

Parameter	True	Mean	SD	2.5%	Median	97.5%
σ_1	0.2000	0.1202	0.0819	0.0316	0.0868	0.3748
σ_2	0.2800	0.2284	0.0624	0.1350	0.2106	0.3766
ρ	0.4500	-0.3303	0.6586	-0.9966	-0.6065	0.7778

Table 8: Posterior summary for the Black-Scholes Margrabe experiment.

Table 8 shows weak recovery of the individual Margrabe parameters. The posterior for ρ is especially disperse, with a credible interval covering almost the entire admissible

region. As a result, in this instance, the exchange option prices do not contain enough information to separately identify both marginal volatilities and the correlation.

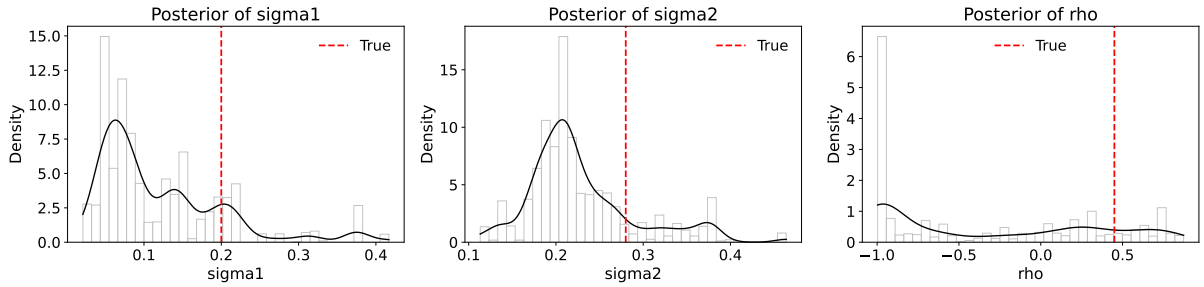


Figure 9: Posterior marginal densities for the Black-Scholes Margrabe experiment. Dashed red lines denote true parameter values.

Figure 9 further illustrates the weak-identification problem. The posterior for σ_1 places considerable mass below the true value. The posterior for σ_2 is closer but still broad, whereas the posterior for ρ is highly dispersed and places substantial mass far from the true value.

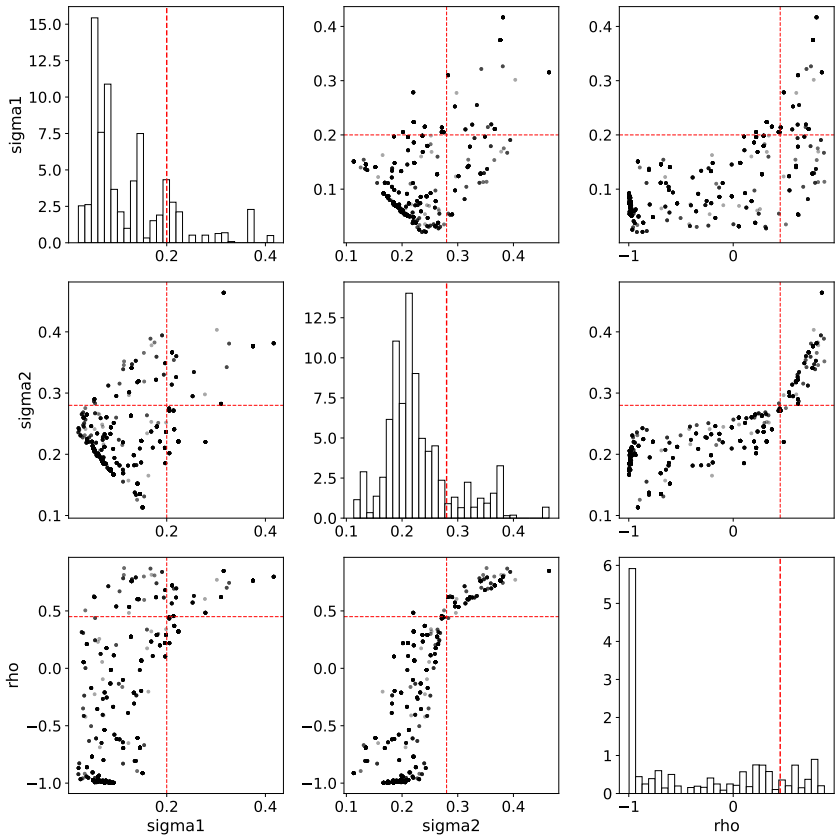


Figure 10: Pairwise posterior dependence for the Margrabe experiment. Dashed red lines denote true parameter values.

We can further analyse the poor mixing by looking at the posterior geometry explicitly, shown in Figure 10. The off-diagonal panels show strong nonlinear dependence between the parameters. This can be explained by the effective volatility of the option $\sigma_{\text{ex}} =$

$\sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$, since σ_{ex} can remain approximately stable while σ_1, σ_2, and ρ move jointly. As a result, the posterior forms a ridge rather than a compact elliptical cloud. In a risk system, we should not calibrate the Margrabe model and then treat the posterior mean of ρ as a precise estimate of correlation. Instead, the posterior dependence should be propagated into downstream prices and risk measures. Alternatively, we could calibrate directly in terms of σ_{ex} if the objective is only to price exchange options, not to infer the underlying covariance structure.

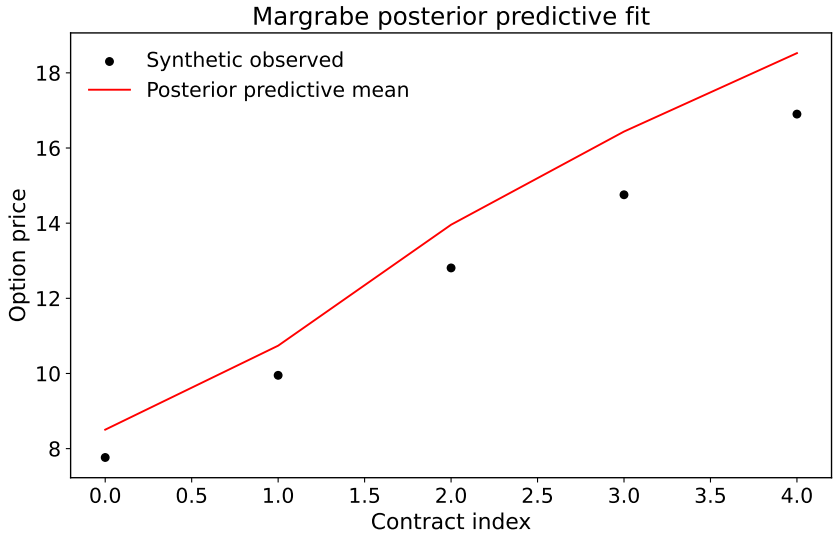


Figure 11: Posterior predictive fit for the Black-Scholes Margrabe experiment.

Figure 11 shows that the posterior predictive mean overprices the synthetic Margrabe observations across the contract index, which is consistent with the parameter-level diagnostics.

5.3 Merton Jump-Diffusion Geometric Asian Option

The Merton Asian experiment is the first main exotic option calibration exercise. Synthetic geometric Asian option prices are generated under the Merton jump-diffusion model and calibrated using the same model. The risk-neutral dynamics are

$$\frac{dS_t}{S_{t^-}} = (r - \lambda\kappa_J)\,dt + \sigma\,dW_t + (J-1)\,dN_t,$$

where $J = e^Y$, $Y \sim \mathcal{N}(\nu, \delta^2)$, and $\kappa_J = \exp\left(\nu + \frac{1}{2}\delta^2\right) - 1$. We can take the calibrated parameter vector as $\boldsymbol{\theta}_{\text{MJD}} = (\sigma, \lambda, \nu, \delta)$.

The sampler uses a Metropolis-Hastings within Gibbs structure, updating the diffusion volatility separately from the jump parameter block.

Parameter	True	Mean	SD	2.5%	Median	97.5%
σ	0.1800	0.1880	0.0078	0.1717	0.1885	0.2017
λ	0.8000	0.4564	0.1787	0.1467	0.4645	0.8089
ν	-0.0400	0.0364	0.0558	-0.0650	0.0334	0.1523
δ	0.1200	0.0848	0.0288	0.0391	0.0820	0.1474

Table 9: Posterior summary for the correctly specified Merton Asian experiment.

Table 9 shows that the diffusion volatility is recovered reasonably well, while the jump parameters are less well recovered. The true value of λ lies near the upper edge of the credible interval, and the posterior for δ is shifted below the true value. The posterior for ν is broad and crosses zero.

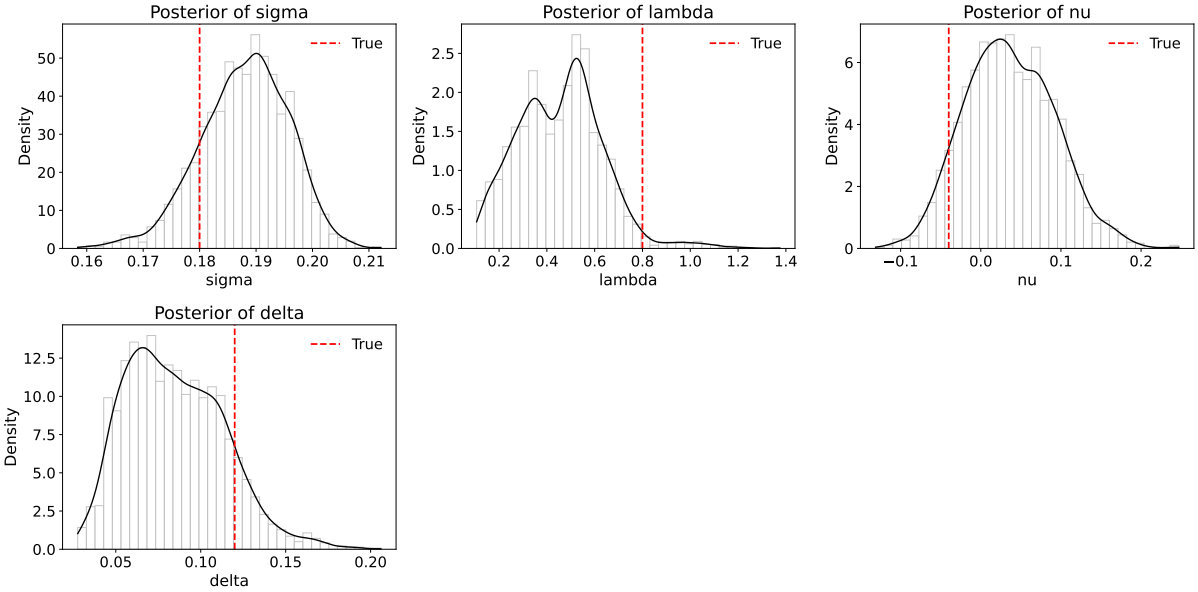


Figure 12: Posterior marginal densities for the correctly specified Merton Asian experiment. Dashed red lines denote true parameter values.

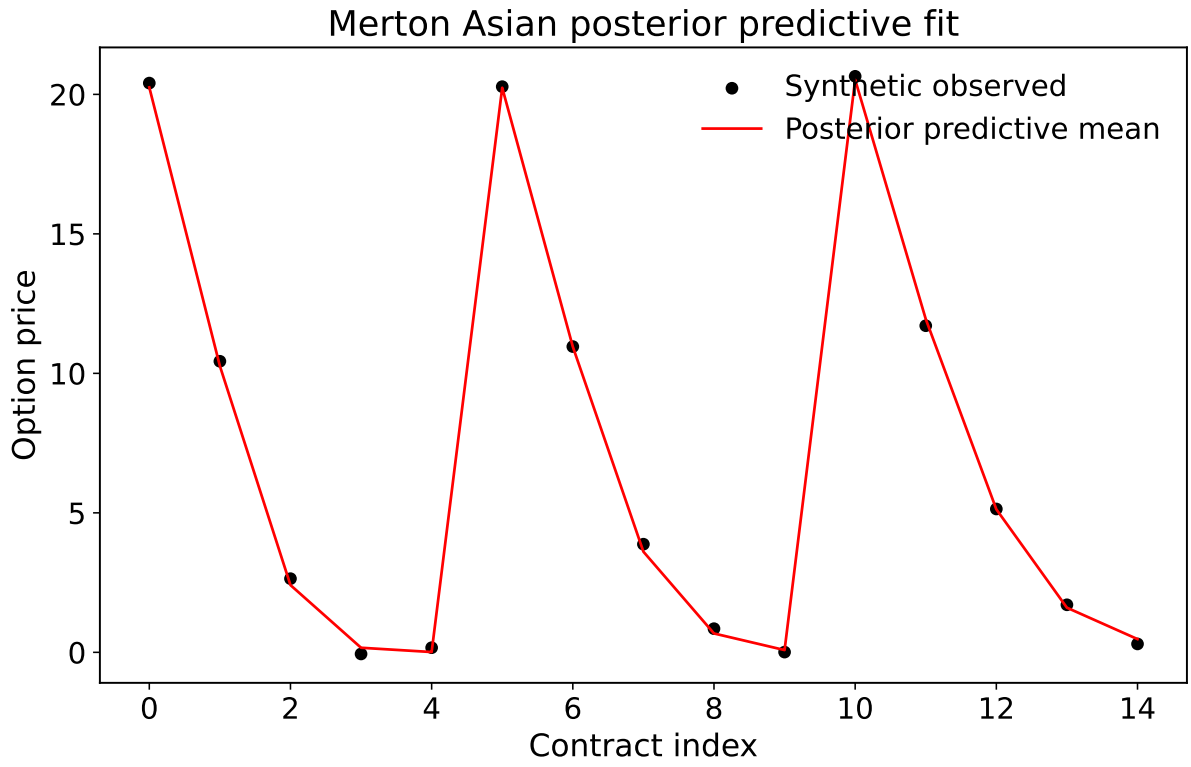


Figure 13: Posterior predictive fit for the correctly specified Merton Asian experiment.

Figure 12 illustrates this weak identification. The posterior for σ is relatively concentrated, while the jump-parameter posteriors are wider and shifted. This means that the option surface identifies the combined distributional effect of jumps more strongly than it identifies the individual jump parameters.

5.4 Heston Geometric Asian Option

The Heston experiment studies stochastic-volatility calibration using synthetic geometric Asian option prices. The data-generating and fitted model are both Heston.

Hamiltonian Monte Carlo is used because the posterior is continuous and strongly dependent. In particular, the parameters v_0, κ, and ξ jointly determine the behaviour of the variance path over the option horizon, while σ and ρ influence the dispersion and asymmetry of the risk-neutral distribution.

Parameter	True	Mean	SD	2.5%	Median	97.5%
v_0	0.0400	0.0320	0.0047	0.0230	0.0317	0.0415
κ	2.0000	2.6534	1.2810	0.9135	2.4764	6.1190
ξ	0.0400	0.0491	0.0127	0.0276	0.0473	0.0764
σ	0.3500	0.3358	0.1323	0.1490	0.3119	0.6206
ρ	-0.5500	-0.4090	0.1556	-0.7300	-0.4031	-0.1256

Table 10: Posterior summary for the correctly specified Heston Asian experiment.

Table 10 shows that the true values generally lie inside the credible intervals, but some

intervals are wide. Most notably, the posterior for κ is especially uncertain, and the posterior for ξ is also broad, meaning that the Asian option surface identifies the broad stochastic-volatility effect more strongly than it identifies each variance-process parameter separately. Figure 14 confirms that parameter mixing is uneven. The posterior for v_0 is concentrated but shifted downward and the posterior for κ is wide and right-skewed. The posterior for ρ remains in the negative-correlation region but is not centred exactly on the true value.

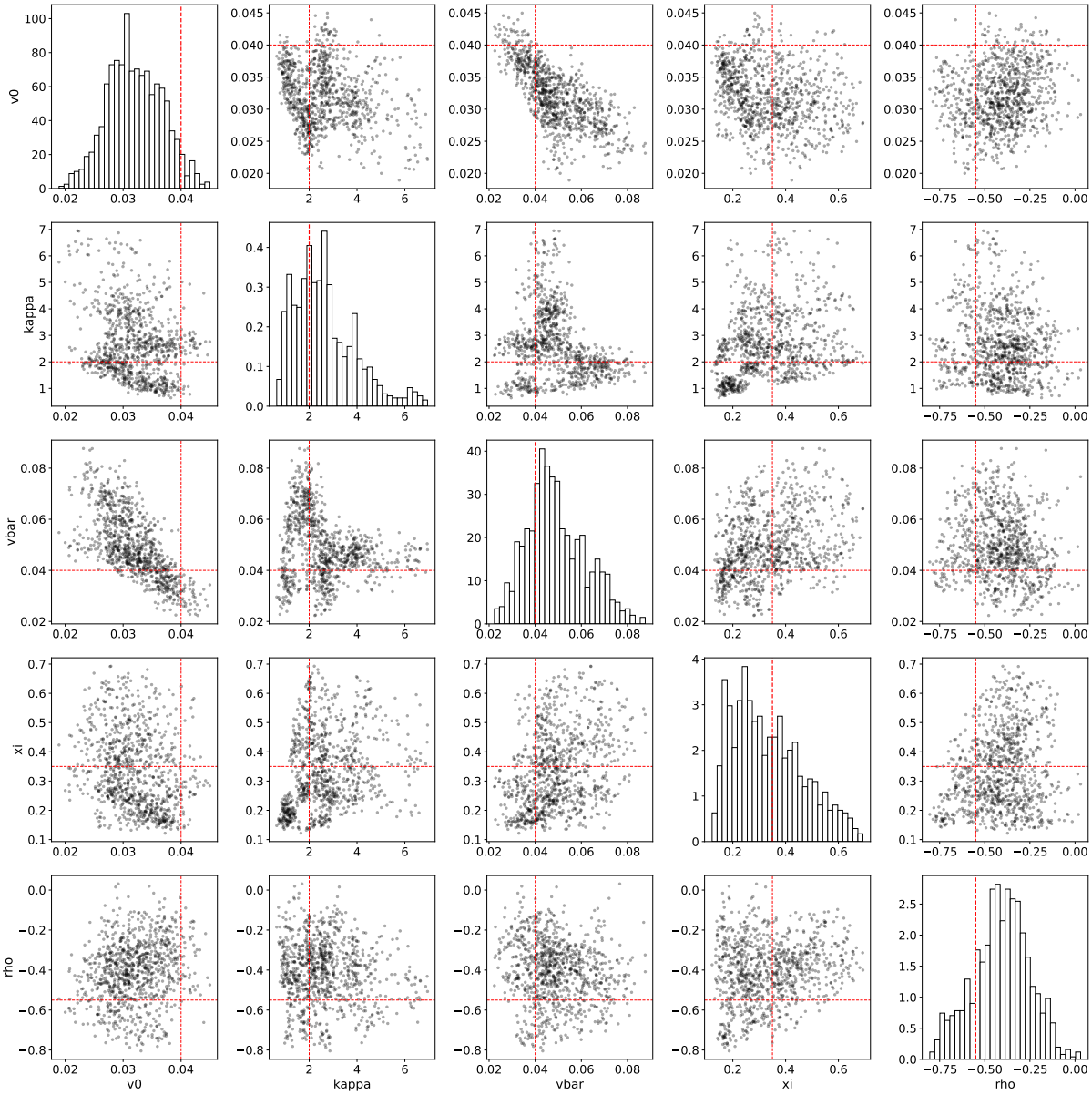


Figure 14: Pairwise posterior dependence in the Heston Asian experiment. Dashed red lines denote true parameter values.

Additionally, Figure 14 shows clear dependence between variance-process parameters. For instance, changes in v_0, κ, and ξ can partly offset each other when calculating the average variance over the option horizon. This is helpful because it shows which parameters should not be interpreted independently. The pairwise posterior structure should be propagated into pricing, hedging, and stress testing. In particular, uncertainty in variance

dynamics can be more important for path-dependent contracts than for the contracts used in calibration.

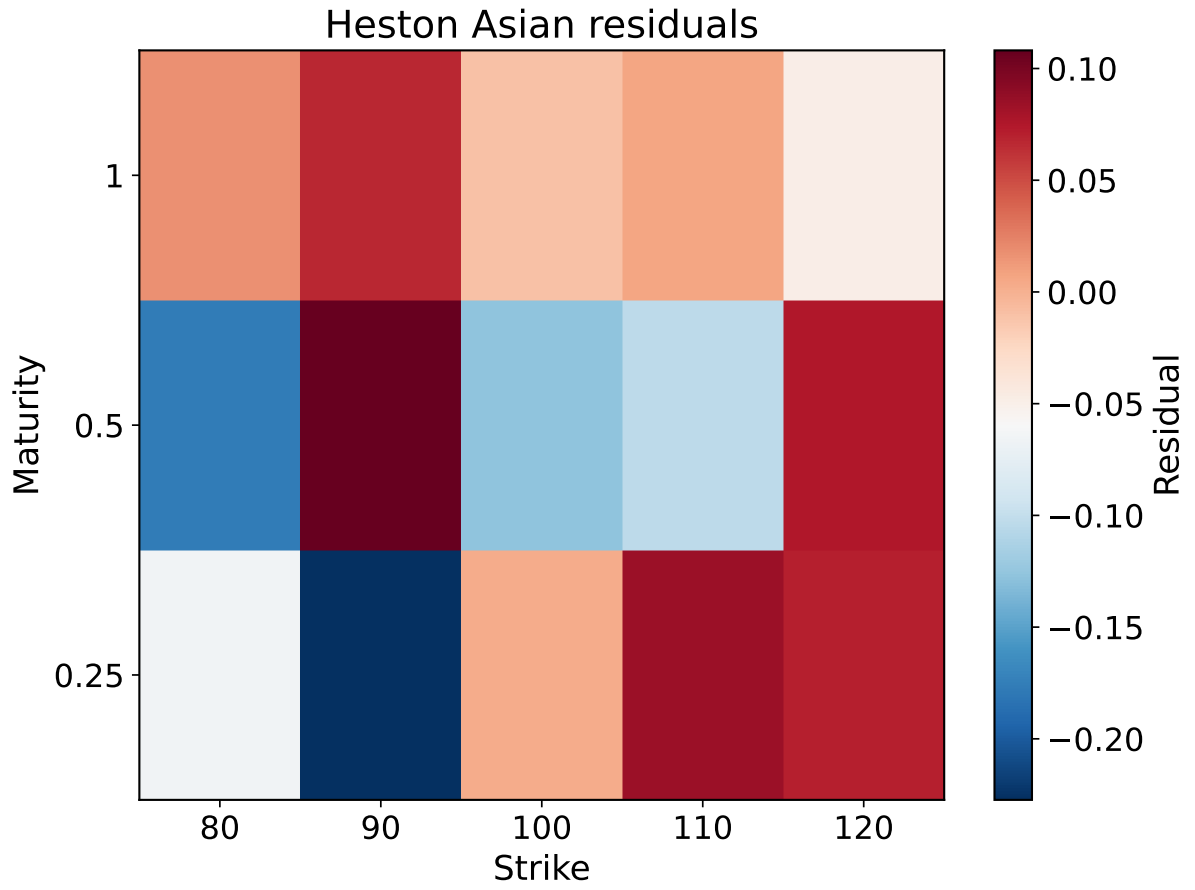


Figure 15: Residual heatmap for the correctly specified Heston Asian experiment.

Figure 15 shows small residuals across the strike-maturity grid. Some local structure remains, but the magnitude is mostly modest. These residuals are most naturally attributed to finite Monte Carlo pricing error, synthetic observation noise, and posterior sampling error.

5.5 Misspecification: Merton Fit to Heston-Generated Asian Prices

The final experiment deliberately fits the wrong model. Synthetic Asian option prices are generated under the Heston model, but the fitted model is Merton jump-diffusion. This experiment is designed to study model risk.

Parameter	Mean	SD	2.5%	Median	97.5%
σ	0.1174	0.0186	0.0720	0.1189	0.1508
λ	1.7226	0.4421	0.9739	1.6750	2.7608
ν	-0.0567	0.0240	-0.1187	-0.0528	-0.0180
δ	0.1003	0.0186	0.0601	0.1010	0.1359

Table 11: Posterior summary for the misspecified Merton fit to Heston-generated Asian prices.

Table 11 shows the fitted Merton model favours a lower diffusion volatility, a relatively high jump intensity, a negative jump mean, and moderate jump volatility. Economically, this suggests that the jump component is being used to mimic some of the asymmetry and dispersion generated by stochastic volatility in the Heston data.

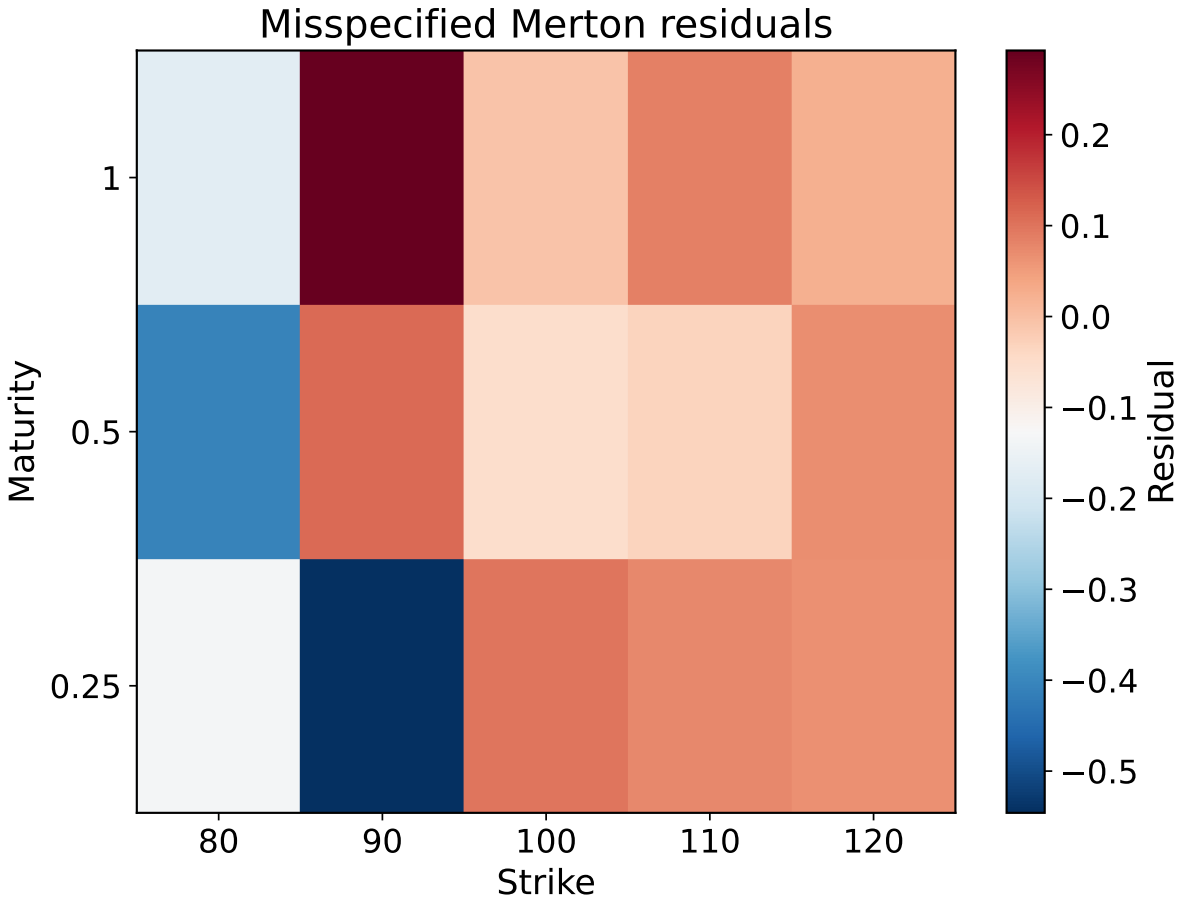


Figure 16: Residual heatmap for the misspecified Merton fit to Heston-generated Asian prices.

Figure 16 shows that the residuals are larger and more structured than in the correctly specified Heston experiment. The overpricing of the options indicates that the Merton model captures the broad level of the surface but fails to reproduce its detailed structure uniformly, which is to be expected.

5.6 Cross-Experiment Comparison

Table 12 summarises sampler efficiency and predictive accuracy across experiments. The minimum effective sample size is noted because the reliability of posterior inference is constrained by the weakest-mixing parameter. Effective sample size per second adjusts this quantity for computational cost. RMSE and MAE summarise posterior predictive accuracy.

Experiment	Runtime	Min ESS	ESS/sec	Acceptance	RMSE	MAE
BS Asian RWM	0.98	1071.51	1094.03	0.491	0.127	0.106
Margrabe Adaptive MH	2.62	6.55	2.50	0.044	1.261	1.197
Birth-death Merton Path	17.65	8.44	0.478	0.185	–	–
Merton Asian MWG	4947.33	24.04	0.0049	0.630	0.163	0.142
Heston Asian HMC	8062.84	4.98	0.0006	0.997	0.100	0.079
Merton fit to Heston data	8575.46	12.64	0.0015	0.510	0.207	0.145

Table 12: Sampler and predictive comparison across experiments.

The Black-Scholes Asian experiment is extremely efficient because it is one-dimensional and analytically priced. The Margrabe experiment is also analytically priced, but its posterior geometry is difficult, leading to poor effective sample size and low acceptance. The Merton and Heston Asian experiments are far more computationally expensive because model-implied prices are simulation-based and the posterior dimension is larger.

The Heston experiment has the lowest RMSE among the Asian option experiments, but it also has very low ESS/sec. The high HMC acceptance rate tells us that the HMC step size may be conservative, i.e., proposals are accepted, but the chain does not move far enough through the posterior per unit time.

The misspecified Merton fit has worse RMSE than the correctly specified Heston fit and a visibly more structured residual heatmap. This is expected because the Merton approximation can match the broad shape of the Heston surface, but it cannot reproduce the surface as uniformly as the correctly specified Heston model.

6 Conclusion

The paper first introduced the relevant option payoffs and stochastic models. The Black-Scholes model was used as an analytically tractable benchmark, while the Merton jump-diffusion and Heston stochastic volatility models were introduced to capture richer features of asset dynamics. The Merton model allows for discontinuous jumps and heavier tails, whereas the Heston model allows volatility itself to evolve randomly over time. These models are more realistic than constant-volatility Black-Scholes dynamics, but they also make calibration more difficult, particularly when the payoff is path-dependent. The geometric Asian option remains tractable because the logarithm of the geometric average is normally distributed under Black-Scholes dynamics. The Margrabe option remains tractable because the exchange ratio can be priced using a change of numeraire.

The Bayesian formulation was then used to move beyond point calibration. Instead of estimating a single parameter vector, the Bayesian approach produces a posterior distribution over plausible parameter values. Several different combinations of volatility, jump, correlation, or stochastic-variance parameters may produce similar option prices. A classical point estimate can hide this uncertainty, while a posterior distribution makes it visible.

The numerical experiments confirmed this distinction. The Black-Scholes geometric Asian experiment behaved as expected with the posterior recovering the true volatility and the posterior predictive surface matching the synthetic prices closely. The Margrabe experiment showed that analytical pricing does not necessarily imply easy calibration. The posterior over the two volatilities and correlation was weakly identified because the option price depends mainly on their combined exchange-volatility effect rather than on each parameter separately.

The birth-death MCMC experiment showed the difficulty of latent jump inference. The sampler identified the most prominent jump in the synthetic Merton path, but smaller jumps were harder to recover. This is a realistic result. In discretely observed data, a small jump can be difficult to distinguish from an unusually large diffusion shock. Therefore, jump inference should be interpreted probabilistically rather than as exact event classification.

Overall, the results support the view that Bayesian MCMC methods are valuable in exotic option pricing because they quantify uncertainty. They reveal when parameters are well identified, when parameters are weakly identified, when posterior dependence is strong, and when a model is misspecified. The main limitation of the approach is computational cost. Embedding Monte Carlo pricing inside MCMC is expensive because each likelihood evaluation may require many simulated paths. This cost becomes more severe for path-dependent contracts, where the whole monitored path must be simulated and stored or approximated.

For this reason, one direction for future work is surrogate modelling, where the option-pricing map is approximated by a faster emulator. In particular, Gaussian process emulators and polynomial chaos expansions could be used to replace repeated expensive pricing calls with cheaper approximations.

A second important direction is dimension reduction for path-dependent payoffs. Instead of representing the entire path through every monitoring date, one can approximate

the path using a lower-dimensional representation that preserves most of the relevant payoff information. As a result, one could look into Brownian bridge constructions and principal-component representations of Brownian paths as possible ways to reduce the effective dimension of the pricing problem, which is directly relevant to the curse of dimensionality caused by path-dependence.

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